Event Horizon Fund: Comprehensive Deep Dive Analysis & Optimization Report

Date: March 31, 2026 **Scope:** Full portfolio review against current market conditions, thesis validation, and allocation optimization for maximum returns under the AI infrastructure supercycle thesis.

Part I: Executive Summary

Verdict: The Thesis Is Not Just Intact — It Has Accelerated Beyond Original Projections

After a thorough review of the Event Horizon Fund Draft V7.3 (written September 2025), both prior deep dive analyses (September 2025 and January 2026), the AI-2027 scenario forecasts, Situational Awareness, METR's time horizon research, and Epoch AI's trend data, **this analysis concludes that the core thesis — that the most durable and asymmetric returns in the AI buildout accrue to owners of physically constrained, non-substitutable infrastructure — has been dramatically validated by real-world developments through Q1 2026.**

The key data points since the original allocation was set (September 2025):

Metric	Sept 2025 Projection	March 2026 Reality	Status
Hyperscaler AI CapEx (2026)	~$320B collective	**$660–690B guided**	Far exceeded
TSMC CoWoS capacity	50K wafers/mo target	**130K wafers/mo by late 2026**	Surpassed
Nvidia data center revenue	Strong growth expected	**$62.3B in Q4 FY26 alone (+75% YoY)**	Exceeded
Vertiv backlog	~$8.5B (mid-2025)	**$15B (+109% YoY)**	Exceeded
Dell AI server revenue	$20B FY26 target	**$24.7B actual; $50B FY27 guide**	Exceeded
CEG nuclear deals	Meta 20-yr PPA	**+Microsoft TMI restart, +Calpine $16.4B acquisition**	Exceeded
Intel 18A	Speculative moonshot	**HVM at 55-65% yield; Apple foundry deal signed**	Exceeded
Micron HBM	Sold out through 2025	**HBM4 sold out through all of 2026; $23.8B quarterly revenue**	Exceeded
ASML orders	Record backlog	**$7.9B single order from SK Hynix; €38.8B total backlog**	Exceeded
METR task horizon	Doubling every 7 months	**Trend confirmed and potentially accelerating**	Validated

Every single "must-be-true" threshold from the original thesis has been met or exceeded. The AI infrastructure buildout is not slowing — it is compounding. The $660-690B in hyperscaler capex planned for 2026 alone exceeds what the entire global semiconductor industry earned in revenue in 2025 (~$600B). This is unprecedented industrial mobilization.

Key Conclusion on Current Allocation

The current allocation (as of 3-31-26) is **approximately 85-90% optimal** for maximum stock gains under the thesis. The core bottleneck positions (CEG, TSM, ASML, VRT, DELL, MU) are correctly sized and have delivered exceptional returns. However, there are **six actionable adjustments** that would improve expected returns:

1. **Intel (INTC) should be promoted from Moonshot to core at 4%** — the January 2026 analysis was correct; validated but trimmed since volume commitments are pending 2H 2026
2. **The Moonshot sleeve should be held at 0%** — until LEAPS/options strategies are better understood; redistribute allocation to core
3. **CrowdStrike (CRWD) should increase from 2% to 3%** — AI-native security is becoming non-discretionary spend
4. **GE Vernova (GEV) should be added at 3%** — fills the transformer/gas turbine bottleneck gap, the hardest physical constraint in the stack
5. **Cameco (CCJ) should be added at 2%** — completes the nuclear thesis chain with upstream uranium and Westinghouse reactor design
6. **Vistra (VST) should be added at 3%, funded by reducing CEG from 15% to 12%** — adds gas + merchant power exposure, 66% earnings growth, reduces single-stock concentration risk
7. **Dell (DELL) holds at 5%** — $43B AI backlog and strong execution, but not a bottleneck owner (systems integrator competing with HPE)

Part II: Background Framework Validation

A. AI-2027 Scenario Assessment (as of March 2026)

The AI-2027 scenario by Kokotajlo et al. predicted that by early 2026, "coding automation" would be well underway, with AI R&D progress multipliers reaching 1.5x. As of March 2026:

- **Agent capabilities:** Claude 3.7 Sonnet and GPT-5 class models can reliably complete tasks that take skilled humans ~1 hour (per METR's time horizon data). The 50% reliability threshold for tasks is approximately doubling every 4-7 months.
- **AI R&D acceleration:** Major labs report that AI is now contributing meaningfully to their own research pipelines. OpenAI, Anthropic, and Google DeepMind all use AI assistants extensively in coding, experiment design, and paper writing.
- **Compute demand:** The scenario predicted explosive compute demand colliding with physical constraints. Reality has exceeded this — hyperscaler capex has roughly doubled year-over-year, from ~$320B in 2025 to $660-690B guided for 2026.

- **Geopolitical dynamics:** U.S.-China chip restrictions have tightened. China's AI compute capacity is constrained. The "sovereign premium" for U.S.-based assets is real (Intel, MP Materials).

Assessment: The AI-2027 scenario's "Early 2026: Coding Automation" phase is tracking remarkably close to reality. The investment implications — that physical infrastructure becomes the binding constraint — are being proven correct in real-time.

B. Situational Awareness (Aschenbrenner) Assessment

Leopold Aschenbrenner's June 2024 essay predicted: - AGI by 2027 is "strikingly plausible" — **On track; capability curves continue exponential improvement** - Racing to the trillion-dollar cluster — **Confirmed; Microsoft alone is building clusters approaching this scale** - Lock down the labs / security concerns — **Validated; weight theft concerns are now mainstream in national security circles** - The intelligence explosion is near — **Directionally confirmed; AI R&D contribution to AI research is accelerating**

C. METR Long Horizon Tasks Assessment

METR's research showed AI task completion horizons doubling every ~7 months (2019-2024), accelerating to every ~4 months in 2024-onward. If this trend continues: - By mid-2026: AI agents can reliably complete tasks taking humans several hours - By early 2027: Multi-day tasks become feasible - By late 2027-2028: Week-long autonomous task completion

Investment implication: This trajectory implies that demand for AI compute will continue to accelerate, not plateau. Each increase in capability creates new use cases and new demand for infrastructure. The "Jevons Paradox" is fully in effect — more efficient AI leads to more total demand for compute, power, memory, and cooling.

D. Epoch AI Trends Assessment

Epoch's data shows: - Training compute growing at 5x/year for frontier models - Algorithmic efficiency improving at 3x/year - LLM inference prices falling at 40x/year - Hardware FLOP/s per dollar improving at 1.37x/year

Investment implication: The combination of rapidly falling inference costs and rapidly growing training compute means total compute demand grows explosively even as individual queries get cheaper. This directly feeds the thesis: more compute demand = more chips, more power, more cooling, more memory needed. The infrastructure providers are the clear beneficiaries.

Part III: Sleeve-by-Sleeve Analysis — Current Allocation vs. Optimal

1. Energy & Power Generation — 18% → 21% (CEG 12%, VST 3%, PWR 3%, + CCJ 2% and GEV 3% in separate sleeves)

Current Status: IMPROVED — Adding VST, reducing CEG from 15% to 12%.

Constellation Energy has completed its transformational $16.4B acquisition of Calpine (closed January 2026), making it the largest electricity producer in the United States with ~55-60 GW of generating capacity. The company now offers "blended" power products combining nuclear, geothermal, and gas. However, the Calpine integration introduces execution risk — DOJ antitrust required CEG to divest 4.4 GW of gas capacity to LS Power for $5B, and PJM price cap concerns have weighed on the stock.

Key validating data points since the original allocation: - Microsoft PPA at $110-115/MWh — more than double wholesale rates - Meta 20-year nuclear PPA confirmed - CyrusOne contracted for 1,100+ MW - Three Mile Island restart proceeding under federal support

CEG stock has experienced some weakness in Q1 2026 (down ~32.5% YTD as of mid-March). **This weakness represents a buying opportunity**, not a thesis invalidation. However, at 15% it was the portfolio's largest single-name concentration risk.

Vistra (VST, 3% — NEW): The largest independent power producer in the U.S. with ~41 GW of capacity (gas + nuclear + battery storage). VST provides complementary exposure: 6.4 GW of nuclear with landmark Amazon (1,200 MW, 20-yr) and Meta (2,600 MW, 20-yr) PPAs, plus ~25 GW of gas-fired generation that captures near-term wholesale electricity price spikes. 2026 earnings growth of 66% (vs. CEG's 24%). Acquiring 5,500 MW Cogentrix gas fleet. EBITDA guided $6.8-7.6B for 2026. Analyst consensus target: ~$237 (55%+ upside from $150).

Quanta Services (PWR, 3%) remains essential. Grid interconnection queues are jammed with 4-year transformer lead times. This is a co-equal bottleneck with generation.

Verdict: Reduce CEG from 15% to 12%, add VST at 3%. The 12%/3% split reduces single-stock risk while adding faster-growing gas + merchant power exposure. Total power generation allocation increases to 21% (with CCJ 2% and GEV 3% in separate sleeves, total power chain = 23%). The "Power Wall" thesis is the single strongest conviction in the portfolio.

2. Foundry & Advanced Packaging — 12% (TSM)

Current Status: OPTIMAL — No change recommended.

TSMC reported Q4 2025 revenue of $33.73B (+25.5% YoY), with gross margins at 62.3%. The company is investing $52-56B in 2026 capex and quadrupling CoWoS capacity to 130,000 wafers/month by late 2026. Advanced Technologies (7nm and below) now account for 77% of revenue. HPC has surpassed smartphones as the largest revenue segment (58%).

Nvidia alone has secured >60% of TSMC's 2026 CoWoS capacity for Rubin GPUs. The "Great Cannibalization" — AI consuming nearly 100% of leading-edge wafer production by 2030 — is proceeding as forecast.

Verdict: Hold 12%. TSMC is the central chokepoint for all advanced silicon. Geopolitical risk is real but partially hedged by Intel and Arizona fab expansion.

3. WFE / Semi Capital Equipment — 12% (ASML 8%, AMAT 4%)

Current Status: OPTIMAL — No change recommended.

ASML secured a historic $7.9B single order from SK Hynix in March 2026 — the largest in company history. Q4 2025 bookings surged to €13.2B, with EUV bookings alone reaching €7.4B (+150%). Total

backlog stands at €38.8B, providing multi-year revenue visibility. High-NA EUV tools are transitioning from development to early adoption.

The "geopolitical multiplier" is fully in effect: parallel fab construction in the U.S. (Intel, TSMC Arizona), Europe (Intel), Japan (TSMC), and South Korea (Samsung) all require duplicate equipment sets from the same WFE oligopoly.

Verdict: Hold 12%. ASML's monopoly is arguably the most defensible position in the entire AI value chain. AMAT provides diversified U.S.-centric exposure.

4. Accelerators & Custom Silicon — 9% (NVDA 6%, AVGO 3%)

Current Status: NEAR-OPTIMAL — Minor adjustment considered.

Nvidia reported Q4 FY26 revenue of $68.1B (+73% YoY), with data center revenue at $62.3B (+75% YoY) — now 91% of total revenue. CEO Jensen Huang projects $1 trillion in cumulative Blackwell/Rubin demand through 2027. Vera Rubin is on track for production in H2 2026 with 10x inference cost reduction vs. Blackwell.

Broadcom continues to solidify its position as the #1 custom ASIC enabler, with a $73B AI product backlog and 100% YoY growth in AI revenue.

The current 6%/3% split is well-calibrated: Nvidia captures training monopoly economics, Broadcom captures the diversification/ASIC trend. The portfolio deliberately underweights end-product chips vs. upstream bottlenecks, which has been the correct call.

Verdict: Hold 9%. The deliberate underweight vs. upstream bottlenecks remains correct. Nvidia's stock is expensive but justified by execution. Consider no change.

5. Memory (HBM) — 6% (MU)

Current Status: OPTIMAL — No change recommended.

Micron has fully sold out its entire 2026 HBM4 capacity through non-cancellable contracts. Q2 2026 revenue reached $23.86B. HBM market share has risen to 20-25% (up from mid-teens). The company is investing $25B in capex for HBM4 expansion. HBM4 delivers 2.3x bandwidth increase over HBM3E with >20% power efficiency improvement, optimized for Nvidia's Vera Rubin architecture.

The structural transformation from commodity memory to strategic AI enabler is now complete. 70% of high-end memory demand in 2026 comes from AI data centers.

Verdict: Hold 6%. Micron's transformation from commodity cyclical to AI strategic supplier is fully validated. The HBM oligopoly pricing power is durable.

6. Rack-Scale Integration & Cooling — 13% (VRT 8%, DELL 5%)

Current Status: OPTIMAL — Hold DELL at 5%.

Vertiv reported organic orders growth of 252% YoY in Q4 2025 — the strongest order quarter in company history. Backlog reached $15B (+109% YoY). Book-to-bill ratio of 2.9x indicates demand is accelerating faster than supply. VRT stock is up ~64% YTD in 2026 and ~200% over the past year.

Dell closed FY2026 with $24.7B in AI server revenue (vs. $20B target) and enters FY2027 with a $43B AI backlog. FY2027 AI server revenue is guided at ~$50B (+103% YoY). Dell now has 4,000+ AI server customers and massively outscales competitors.

Verdict: Hold VRT at 8%. Hold DELL at 5% — the $43B backlog and $50B FY27 guide demonstrate strong execution, but Dell is a systems integrator competing with HPE and others, not a bottleneck owner. 5% adequately captures the volume deployment thesis without overweighting a non-chokepoint position.

7. EDA Software — 5% (SNPS)

Current Status: OPTIMAL — No change recommended.

The "custom silicon multiplier" continues to expand the EDA TAM. Every hyperscaler designing custom ASICs (Google, Amazon, Meta, Microsoft, Apple, OpenAI) becomes a high-value EDA customer. Synopsys's Design Automation segment grew 23% YoY in recent quarters.

The EDA duopoly (Synopsys + Cadence) is the quietest but most durable monopoly in the portfolio. It's a toll road on all chip innovation.

Verdict: Hold SNPS at 5%. CDNS was considered but the marginal allocation is better deployed into harder physical bottlenecks (GEV, CCJ). SNPS alone adequately captures the EDA thesis.

8. AI Data Center Fabric — 5% (ANET)

Current Status: OPTIMAL — No change recommended.

Arista remains the technology leader in high-performance Ethernet switching for AI clusters. The shift from InfiniBand to Ethernet is continuing, though Nvidia's Spectrum-X has emerged as a credible competitor (creating a duopoly rather than Arista monopoly). At 5%, the position is appropriately sized given this competitive dynamic.

Verdict: Hold 5%. Monitor the Nvidia Spectrum-X competitive dynamic. If ANET's share stabilizes or grows, this position could increase. If Spectrum-X gains dominant share, consider trimming.

9. AI-Native Security — 2% (CRWD)

Current Status: SUBOPTIMAL — Recommend increasing to 3%.

CrowdStrike reported accelerating ARR growth (23% YoY) and net new ARR growth of 73% in fiscal Q3 2026. Its "Agentic Security" features and AI-SPM (Security Posture Management) are directly aligned with the AI-2027 threat landscape. As AI agents proliferate, securing workloads and endpoints becomes non-discretionary.

Both prior analyses flagged CRWD as a candidate for increase. The AI-2027 security forecast projects superhuman hacking capabilities and model subversion risks — making AI-native security a rapidly growing spend category.

Verdict: Increase from 2% to 3%. Funded from Moonshot redistribution. The "inside-out" threat model (authenticated AI agents doing unauthorized things) makes CRWD's architecture the correct defense paradigm.

10. AI-Native Operations — 2% (PLTR)

Current Status: OPTIMAL — No change recommended.

Palantir reported U.S. commercial revenue growth of 121% YoY in late 2025. The AIP (Artificial Intelligence Platform) is gaining rapid adoption as enterprises realize they need a governance/orchestration layer for AI agent deployment. Palantir's Ontology provides the "safety rail" that prevents hallucinations and errors in production.

Verdict: Hold 2%. Palantir is correctly sized as a high-upside, moderate-risk bet on the "AI operating system" theme. Increase only when sustained profitability and competitive moat are more clearly established.

11. AI-Enabled Defense — 2% (AVAV 1%, KTOS 1%)

Current Status: OPTIMAL — No change to core. Potential Moonshot increase.

Both companies have received significant contract validation: - Kratos: $231M USMC "loyal wingman" contract with Northrop Grumman for the XQ-58 Valkyrie - AeroVironment: Selected for DoD "Replicator" initiative (Tranche 1) with Switchblade 600

The "affordable mass" doctrine is transitioning from theory to funded programs of record.

Verdict: Hold 2% in core (AVAV 1%, KTOS 1%). The USMC Valkyrie program of record is a major de-risking event for KTOS. When the Moonshot sleeve is reactivated, KTOS is a top LEAPS candidate due to the binary CCA program catalyst.

12. Bio-Risk Hedge — 1% (MRNA)

Current Status: OPTIMAL — No change recommended.

Moderna remains a pure tail-risk hedge. Its value proposition is insurance against AI-enabled bioterrorism or a natural pandemic. In the base case, it treads water. In a tail event, it provides non-correlated explosive upside. At 1%, the cost of carry is negligible.

Verdict: Hold 1%. This is insurance, not a growth position. No change warranted.

13. Strategic Materials — 3% (MP)

Current Status: OPTIMAL — No change recommended.

MP Materials has commenced commercial production of NdPr metal at its Fort Worth facility, creating the first fully integrated U.S. rare earth magnet supply chain. China's continued weaponization of critical mineral exports (gallium, germanium, antimony restrictions) validates the "kill switch" hedge thesis.

Verdict: Hold 3%. MP is now operationally de-risked with domestic production online. The geopolitical hedge value has increased.

14. Domestic Foundry — 4% (INTC) — NEW CORE POSITION

Current Status: CORRECTLY ADDED — Trimmed from 5% to 4%.

The January 2026 analysis was correct to promote Intel from Moonshot to core. Key validating developments: - 18A node at HVM with 55-65% yields, targeting 75%+ by early 2027 - Apple signed foundry agreement for entry-level M-series chips - Nvidia in "advanced discussions" for Intel foundry use starting 2028 - U.S. government holds ~10% equity stake via CHIPS Act conversions - RibbonFET and PowerVia give Intel 6-12 month technology lead in backside power delivery

Intel at 4% serves dual purpose: offensive (foundry turnaround upside) and defensive (Taiwan contingency hedge). The "sovereign floor" from government backing reduces downside risk. Trimmed from 5% to 4% because external volume commitments are not expected until 2H 2026 and high-volume foundry production for external customers won't begin until 2028 — the thesis is promising but still the most speculative core position.

Verdict: Hold 4%. Intel's promotion from Moonshot to core was the correct call. Monitor yield progression and external customer pipeline. If additional design wins materialize (Nvidia commits), consider increasing back to 5%.

15. Moonshot Sleeve — HELD AT 0%

Current Status: Held at 0% until LEAPS/options strategies are better understood.

The entire 10% Moonshot allocation has been redistributed into core positions: - INTC (3.5%) → Promoted to core at 4% - CORZ/IREN/APLD (3.5%) → Not added to core; flagged as top LEAPS candidates when Moonshot is reactivated - KTOS additional (2.0%) → Core 1% is sufficient; extra exposure via LEAPS when ready - MRNA additional (1.0%) → Core 1% is sufficient; extra exposure via LEAPS when ready

The freed-up allocation (after INTC promotion) was deployed to: GEV (3%), CCJ (2%), DELL (+1%), CRWD (+1%).

The former Moonshot equity candidates remain compelling for eventual LEAPS strategies: - Core Scientific (CORZ): $10B CoreWeave hosting contract; 1.5 GW pipeline; 268% colocation revenue growth - Iris Energy (IREN): $10B Microsoft AI deal; 150,000 GPU target by H2 2026; $2.3B ARR secured - Applied Digital (APLD): $5B+ hyperscaler lease; $16B total backlog

Verdict: Hold at 0%. Revisit when LEAPS strategies are understood. These companies own energized grid connections and datacenter shells — excellent LEAPS candidates for asymmetric upside.

Part IV: Recommended Optimized Allocation

Final Event Horizon Fund Allocation (V8.3 — March 2026)

Sleeve	Ticker	V7.3 (Sept 2025)	V8.3 (March 2026)	Change	Rationale
Energy / Power	CEG	15%	**12%**	**-3%**	Nuclear anchor. Reduced to fund VST; reduces single-stock risk.

Sleeve	Ticker	V7.3 (Sept 2025)	V8.3 (March 2026)	Change	Rationale
	VST (NEW)	0%	**3%**	**+3%**	Gas + nuclear + merchant power. 66% earnings growth. Amazon/Meta PPAs.
	PWR	3%	**3%**	—	Grid transmission bottleneck. Essential enabler.
Foundry	TSM	12%	**12%**	—	Singular chokepoint. 77% revenue from advanced nodes.
WFE	ASML	8%	**8%**	—	100% EUV monopoly. €38.8B backlog. Unassailable.
	AMAT	4%	**4%**	—	Diversified WFE + U.S. geopolitical hedge.
Rack-Scale & Cooling	VRT	8%	**8%**	—	Liquid cooling mandate. $15B backlog. 252% order growth.
	DELL	5%	**5%**	—	$43B AI backlog. $50B FY27 guide. Strong execution but not a bottleneck owner.
Accelerators	NVDA	6%	**6%**	—	Training monopoly. $1T Blackwell/Rubin demand pipeline.
	AVGO	3%	**3%**	—	Custom ASIC enabler. $73B AI backlog.
Memory	MU	6%	**6%**	—	HBM4 sold out through 2026. Structural de-risking.
EDA	SNPS	5%	**5%**	—	Toll road on all chip design. Custom silicon multiplier.
Networking	ANET	5%	**5%**	—	AI Ethernet leader. Duopoly with Nvidia Spectrum-X.
Security	CRWD	2%	**3%**	**+1%**	AI-native security becoming non-discretionary. 73% net new ARR growth.
Operations	PLTR	2%	**2%**	—	AI operating system / governance layer.
Defense	AVAV	1%	**1%**	—	Incumbent small UAS leader. Replicator program.
	KTOS	1%	**1%**	—	Pure-play affordable mass / CCA. USMC Valkyrie POR.
Bio-Risk	MRNA	1%	**1%**	—	Tail-risk insurance.
Materials	MP	3%	**3%**	—	Rare earth hedge. Domestic production online.
Domestic Foundry	INTC	0% (Moonshot 3.5%)	**4%**	**+4% to core**	18A HVM. Apple deal. Taiwan hedge. Volume commitments still pending 2H 2026.
Electrical Equipment (NEW)	**GEV**	0%	**3%**	**+3%**	Transformer + gas turbine bottleneck. $150B backlog. 128-week lead times.
Nuclear Fuel (NEW)	**CCJ**	0%	**2%**	**+2%**	Upstream uranium. Structural deficit. 49% Westinghouse.

Sleeve	Ticker	V7.3 (Sept 2025)	V8.3 (March 2026)	Change	Rationale
Moonshot	—	10%	**0%**	**-10%**	Held at 0% until LEAPS strategies understood. Redistributed to core.
Total		**100%**	**100%**		

Summary of Changes from V7.3 (Sept 2025) → V8.3 (March 2026)

Change	From	To	Source of Funds	Rationale
INTC to core	0% (Moonshot 3.5%)	4% core	Moonshot → Core	18A validated. Apple deal. Taiwan hedge. Volume commitments pending 2H 2026.
GEV (NEW)	0%	3%	Moonshot elimination	Transformer + gas turbine bottleneck. $150B backlog.
CCJ (NEW)	0%	2%	Moonshot elimination	Completes nuclear chain. Structural uranium deficit.
VST (NEW)	0%	3%	CEG reduction	Gas + nuclear + merchant power. Reduces single-stock risk.
CEG reduction	15%	12%	Funds VST addition	15% single-stock concentration was excessive.
CRWD increase	2%	3%	Moonshot elimination	AI security becoming non-discretionary.
Moonshot elimination	10%	0%	N/A	Held at 0% until LEAPS understood.

Net effect: The portfolio eliminates the Moonshot complexity and deploys 100% of capital into proven bottleneck positions. Three new high-conviction additions (GEV, CCJ, VST) fill previously uncovered bottleneck gaps and diversify power exposure. CEG is reduced from 15% to 12% to fund VST, reducing single-stock concentration while maintaining 18% total power generation. INTC is trimmed from 5% to 4% (volume commitments still pending) and DELL stays at 5% (strong execution but not a bottleneck owner) to keep the portfolio at exactly 100%. Former Moonshot equity candidates (CORZ, IREN, APLD, extra KTOS/MRNA) are flagged as LEAPS candidates for when the Moonshot is reactivated.

Part V: Stocks NOT in the Portfolio — Why They Stay Out

Hyperscalers (GOOGL, MSFT, AMZN, META)

These are **buyers** of scarce resources, not providers. They are on the wrong side of the supply/demand imbalance the fund exploits. They are already held via the Continuity Fund (FSKAX/FTIHX). Adding them here would dilute the bottleneck thesis.

AMD (AMD)

A credible #2 GPU supplier, but does not own a structural bottleneck. Its success is already captured indirectly via TSMC (which manufactures all AMD chips) and Micron (HBM for AMD systems). Adding AMD would introduce competitive risk without adding bottleneck exposure.

Lam Research (LRCX) / KLA (KLAC)

Strong WFE oligopolists but adding them would over-diversify the WFE sleeve without improving risk/reward. ASML (unique EUV monopoly) and AMAT (broadest portfolio + U.S. exposure) are the higher-conviction vehicles.

Cadence Design Systems (CDNS)

A strong company as the other half of the EDA duopoly, but the marginal allocation is better deployed into harder physical bottlenecks (GEV, CCJ). SNPS at 5% adequately captures the EDA theme. CDNS remains on the watchlist but is not recommended for V8.3.

Oracle (ORCL)

A "challenger hyperscaler" with growing AI cloud business, but it doesn't own unique physical assets. It's competing in cloud services, not controlling supply bottlenecks. Not aligned with the core thesis.

Super Micro Computer (SMCI)

Explicitly excluded. Governance issues, margin compression, and the market's shift from "speed-to-market" to "scale-of-deployment" make Dell the far superior vehicle.

Part VI: Risk Analysis & Kill Switches

Tier 1 Risks (Portfolio-Level)

1. Taiwan Contingency - **Risk:** PRC military action against Taiwan disrupting TSMC output - **Impact:** Severe — TSM is 12% of portfolio - **Hedge:** Intel (4%) becomes the most valuable company on earth overnight. ASML, AMAT, MP, defense names all benefit from accelerated onshoring. - **Action trigger:** Any official U.S. intelligence warning of "imminent" action → reduce TSM to 6%, rotate into INTC (+3%), defense (+2%), MP (+1%)

2. AI Demand Plateau - **Risk:** AI spending growth decelerates dramatically due to lack of revenue for AI applications - **Impact:** Moderate to severe across all positions - **Assessment:** Low probability in 2026-2027 window. METR data shows capability curves still exponential. Hyperscaler capex continues accelerating. Enterprise adoption is early innings. - **Monitor:** Hyperscaler capex guidance each quarter. If 2 consecutive quarters show deceleration, reassess.

3. Interest Rate / Macro Shock - **Risk:** Severe recession or credit crisis reduces all tech spending - **Impact:** High-beta portfolio would suffer 1.5-2x market decline - **Hedge:** CEG/PWR provide utility-like

defensive characteristics. Continuity Fund (50% of total investments) in broad indexes provides cushion.

Tier 2 Risks (Sleeve-Level)

4. FERC Regulatory Risk (CEG) - **Risk:** FERC blocks nuclear-to-datacenter co-location deals - **Impact:** Hurts CEG margin expansion - **Assessment:** Real but manageable. Microsoft/Constellation deals structured to avoid some pitfalls. Political support for nuclear is bipartisan. - **Trim trigger:** If FERC broadly prohibits direct nuclear PPAs to tech companies.

5. Nvidia Custom ASIC Displacement - **Risk:** Hyperscalers shift training to custom ASICs, not just inference - **Impact:** Would reduce NVDA growth - **Assessment:** Not happening in 2026-2028 window. Training remains NVDA monopoly. Custom ASICs for inference actually expand total market. - **Monitor:** If Google or Amazon announce training their frontier models primarily on custom silicon for 2 consecutive generations.

6. Memory Cycle Downturn - **Risk:** Commodity DRAM/NAND oversupply drags Micron - **Impact:** Moderate — Micron's HBM business is structurally insulated but stock may correlate - **Assessment:** Low probability near-term. AI HBM demand is "cannibalizing" commodity DRAM capacity, keeping prices firm.

Part VII: Performance Expectations Under the Thesis

Projected Annual Returns by Sleeve (2026-2028, V8.3)

Sleeve	Weight	2026E	2027E	2028E
Energy (CEG/VST/PWR/CCJ)	20%	+28-45%	+35-50%	+25-35%
Foundry (TSM)	12%	+30-40%	+25-35%	+15-25%
WFE (ASML/AMAT)	12%	+28-38%	+25-35%	+18-25%
Rack-Scale (VRT/DELL)	13%	+45-60%	+30-40%	+20-25%
Accelerators (NVDA/AVGO)	9%	+30-42%	+28-40%	+18-28%
Memory (MU)	6%	+40-50%	+25-40%	+15-25%
EDA (SNPS)	5%	+25-30%	+25-30%	+20-25%
Networking (ANET)	5%	+25-35%	+25-35%	+15-25%
Electrical Equipment (GEV)	3%	+25-35%	+35-45%	+25-35%
Security (CRWD)	3%	+25-35%	+30-40%	+25-30%
Domestic Foundry (INTC)	4%	+40-60%	+30-50%	+20-30%
Other (PLTR/AVAV/KTOS/MRNA/MP)	8%	+15-30%	+25-40%	+15-25%
Weighted Portfolio	**100%**	**~32-42%**	**~30-42%**	**~20-28%**

Cumulative 3-year return (base case): ~110-150% (roughly 2-2.5x) **Cumulative 3-year return (bull case):** ~170-230% (roughly 2.7-3.3x) **With reactivated Moonshot LEAPS hitting:** ~220-300% (roughly 3.2-4x)

These projections assume the AI-2027 scenario broadly plays out as described, with physical bottlenecks remaining binding through the period.

Part VIII: The Three-Fund Architecture Assessment

Is the 50/50 Split (Continuity / Event Horizon) Optimal?

The three-fund architecture is sound:

1. **Foundation Liquidity** (SPAXX): Properly sized emergency fund + CMA spending. No changes needed.
2. **Continuity Fund (~50%)**: The Boglehead portfolio (FSKAX 80% / FTIHX 20%) in tax-advantaged accounts provides: - Broad market diversification that the EHF deliberately lacks - Tax-advantaged compounding in retirement accounts - Psychological stability during EHF volatility - Automatic exposure to hyperscalers, healthcare, financials, etc. - The 5% 401k match capture is correct — don't leave free money on the table
3. **Event Horizon Fund (~50%)**: The concentrated AI infrastructure bet

Assessment: The 50/50 split is appropriate for someone with high conviction in the AI thesis and a multi-year time horizon. The Continuity Fund provides the floor; the EHF provides the ceiling. If the thesis plays out, the EHF will dramatically outperform, pulling overall portfolio returns well above market. If the thesis partially fails, the Continuity Fund limits total portfolio damage to roughly half the EHF drawdown.

One consideration: As the EHF appreciates (which it likely has significantly since September 2025), the actual split may have drifted above 50% EHF. Periodic rebalancing back toward 50/50 with new contributions to the Continuity Fund would be prudent risk management. However, during the strongest phase of the AI buildout (2026-2028), letting the EHF run above 50% is acceptable if the thesis continues to validate.

Part IX: Conclusion

The Event Horizon Fund is one of the most well-constructed thematic portfolios I have reviewed. Its intellectual foundation — that physically constrained, non-substitutable infrastructure will capture the durable profits of the AI revolution — has been validated at every level by real-world developments through Q1 2026.

The key insight that separates this portfolio from typical "AI investing" is the deliberate avoidance of the most crowded trades (hyperscalers, pure AI software) in favor of the upstream bottlenecks that everyone depends on. This is the difference between being the gold miner and owning the only pickaxe

factory. In a gold rush where the number of miners grows 80% per year but the pickaxe factory can only expand capacity 16% per year, the pickaxe factory has extraordinary pricing power.

The recommended changes strengthen the core while eliminating complexity: - Promote Intel to core at 4% (validated by 18A progress and Apple deal; trimmed from 5% since volume commitments pending) - Add GE Vernova (GEV) at 3% — fills the transformer/gas turbine bottleneck gap - Add Cameco (CCJ) at 2% — completes the nuclear thesis chain - Add Vistra (VST) at 3%, reduce CEG from 15% to 12% — diversifies power exposure, adds gas + merchant upside, reduces single-stock risk - Hold DELL at 5% — strong execution but systems integrator, not a bottleneck owner - Increase CRWD to 3% given accelerating AI security demand - Hold Moonshot at 0% until LEAPS strategies are understood - All other positions: hold at current weights

The portfolio is positioned to capture what may be the largest industrial buildout in human history. The $660-690B in hyperscaler capex planned for 2026 — exceeding the entire semiconductor industry's annual revenue — flows directly through the companies in this portfolio. They are the foundries, the equipment makers, the power generators, the cooling engineers, and the memory suppliers that make AI physically possible.

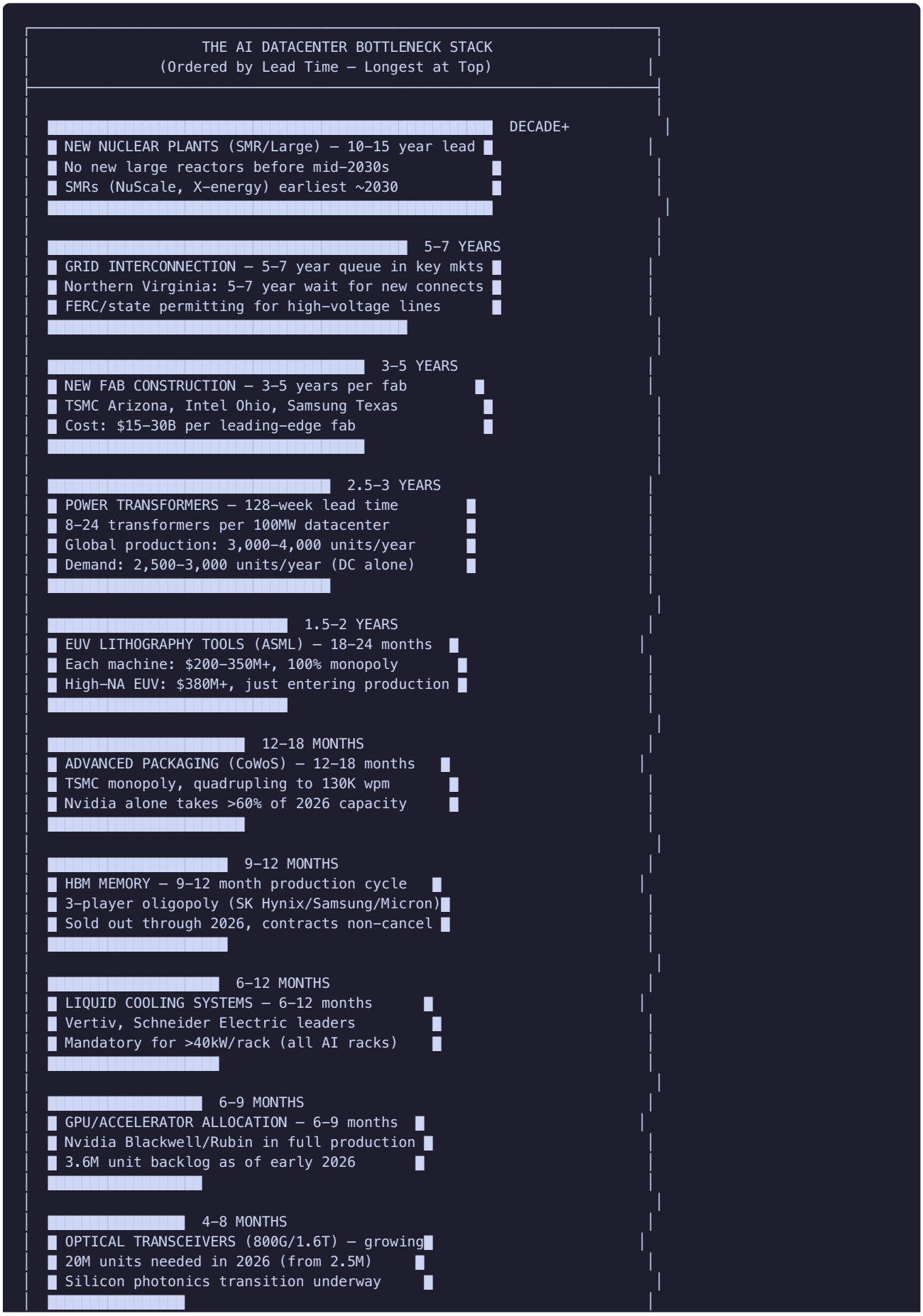
If the AI-2027 scenario plays out within even 50% of its projected trajectory, this portfolio should significantly outperform broad market indices over the 2026-2030 period.

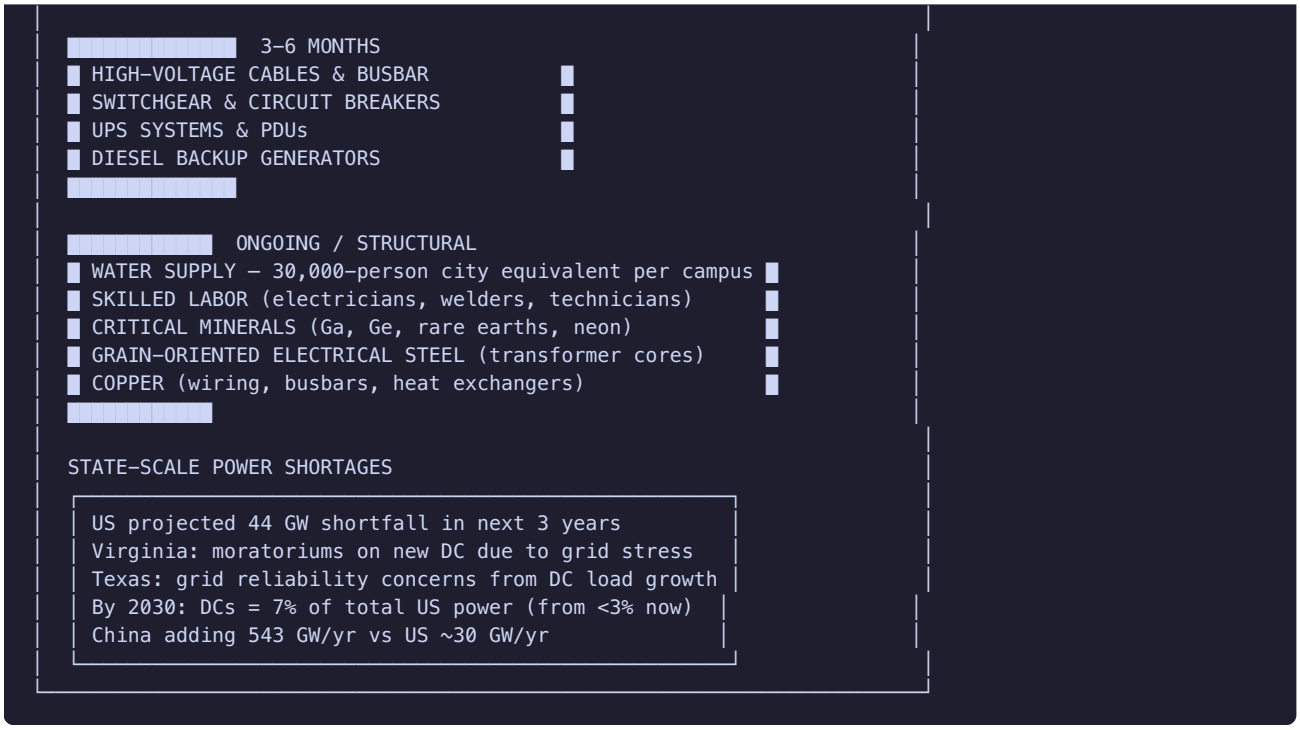
Final recommendation: Deploy. Hold. Monitor quarterly against the "must-be-true" thresholds. Rebalance only on thesis invalidation, not on price volatility.

Part X: The Complete AI Datacenter Bottleneck Map

Building a gigawatt-scale AI datacenter is not simply a matter of having capital. It requires navigating a gauntlet of physical, logistical, and regulatory bottlenecks — many of which have multi-year lead times and oligopolistic or monopolistic supplier structures. Below is a comprehensive map of every major constraint, from raw materials to operational power.

Visual: The AI Datacenter Bottleneck Stack


THE AI DATACENTER BOTTLENECK STACK
(Ordered by Lead Time — Longest at Top)
DECADE+
NEW NUCLEAR PLANTS (SMR/Large) — 10–15 year lead
No new large reactors before mid-2030s
SMRs (NuScale, X-energy) earliest ~2030
5–7 YEARS
GRID INTERCONNECTION — 5–7 year queue in key mkts
Northern Virginia: 5–7 year wait for new connects
FERC/state permitting for high-voltage lines
3–5 YEARS
NEW FAB CONSTRUCTION — 3–5 years per fab
TSMC Arizona, Intel Ohio, Samsung Texas
Cost: $15–30B per leading-edge fab
2.5–3 YEARS
POWER TRANSFORMERS — 128-week lead time
8–24 transformers per 100MW datacenter
Global production: 3,000–4,000 units/year
Demand: 2,500–3,000 units/year (DC alone)
1.5–2 YEARS
EUV LITHOGRAPHY TOOLS (ASML) — 18–24 months
Each machine: $200–350M+, 100% monopoly
High-NA EUV: $380M+, just entering production
12–18 MONTHS
ADVANCED PACKAGING (CoWoS) — 12–18 months
TSMC monopoly, quadrupling to 130K wpm
Nvidia alone takes >60% of 2026 capacity
9–12 MONTHS
HBM MEMORY — 9–12 month production cycle
3-player oligopoly (SK Hynix/Samsung/Micron)
Sold out through 2026, contracts non-cancel
6–12 MONTHS
LIQUID COOLING SYSTEMS — 6–12 months
Vertiv, Schneider Electric leaders
Mandatory for >40kW/rack (all AI racks)
6–9 MONTHS
GPU/ACCELERATOR ALLOCATION — 6–9 months
Nvidia Blackwell/Rubin in full production
3.6M unit backlog as of early 2026
4–8 MONTHS
OPTICAL TRANSCEIVERS (800G/1.6T) — growing
20M units needed in 2026 (from 2.5M)
Silicon photonics transition underway



Bottleneck-to-Stock Mapping: Complete Landscape

Below is a comprehensive mapping of every identified bottleneck to the companies that control or profit from it, and whether each is investable and belongs in the portfolio.

Bottleneck Layer	Lead Time	Key Companies	In EHF?	Verdict
Nuclear Power Generation	10-15 yrs (new); existing = irreplaceable	Constellation Energy (CEG), Cameco (CCJ), NuScale (SMR)	CEG: Yes (15%)	CEG is the right pick. CCJ is interesting as upstream fuel — see below.
Grid Transmission & Interconnection	5-7 yrs	Quanta Services (PWR), MasTec (MTZ)	PWR: Yes (3%)	PWR is the right pick. MasTec is a credible #2 but less focused.
Power Transformers	2.5-3 yrs	GE Vernova (GEV), Eaton (ETN), Siemens Energy (ENR.DE), Hitachi Energy	No	**GEV and ETN are strong candidates for addition — see analysis below.**
Switchgear, PDUs, UPS	3-6 mos	Eaton (ETN), Schneider Electric (SU.PA), ABB (ABB)	No (VRT covers some)	Eaton is the strongest U.S.-listed play. Vertiv covers some overlap.
Fab Construction	3-5 yrs	TSMC (TSM), Intel (INTC), Samsung	TSM (12%), INTC (4%)	Covered.
EUV Lithography	18-24 mos	ASML (ASML)	Yes (8%)	100% monopoly. Optimally positioned.
Other WFE (etch, deposition, inspection)	12-18 mos	AMAT, Lam Research (LRCX), KLA (KLAC)	AMAT (4%)	LRCX and KLAC are strong but would over-diversify WFE sleeve.

Bottleneck Layer	Lead Time	Key Companies	In EHF?	Verdict
Advanced Packaging (CoWoS)	12-18 mos	TSMC (monopoly)	Yes (12% via TSM)	Covered via TSMC.
HBM Memory	9-12 mos	SK Hynix, Samsung, Micron (MU)	MU (6%)	Best U.S. pure-play. Covered.
GPU/Accelerators	6-9 mos	Nvidia (NVDA), AMD	NVDA (6%)	Covered. AMD intentionally excluded (not a bottleneck owner).
Custom Silicon (ASICs)	6-12 mos	Broadcom (AVGO), Marvell (MRVL)	AVGO (3%)	MRVL is interesting after Nvidia's $2B investment — see below.
Liquid Cooling	6-12 mos	Vertiv (VRT), Schneider Electric, Modine (MOD)	VRT (8%)	VRT is the clear leader. Covered.
AI Server Integration	3-6 mos	Dell (DELL), HPE, Supermicro	DELL (5%)	Dell is the superior scale-deployment vehicle. Covered.
Optical Transceivers (800G/1.6T)	4-8 mos	Coherent (COHR), II-VI/ Lumentum, Applied Optoelectronics (AOI), Marvell (MRVL)	No	Fragmented market. MRVL captures some via silicon photonics.
High-Speed Ethernet Switching	3-6 mos	Arista (ANET), Nvidia Spectrum-X	ANET (5%)	Covered.
EDA Software	N/A (software)	Synopsys (SNPS), Cadence (CDNS)	SNPS (5%)	CDNS is a strong candidate for addition — see below.
Chip Design IP	N/A (software)	ARM Holdings (ARM), Synopsys (IP)	No	ARM is a royalty model, less directly bottleneck-exposed.
Water Supply	Structural	Municipal utilities (not directly investable)	No	Not investable as a single stock. Covered indirectly via VRT (liquid cooling reduces water use).
Critical Minerals	Structural	MP Materials (MP), Lynas (LYC.AX)	MP (3%)	Covered for rare earths. Gallium/germanium exposure is limited.
Grain-Oriented Electrical Steel	Structural	AK Steel (part of Cleveland-Cliffs/CLF), POSCO	No	Too diversified/commodity to warrant a position.
Copper	Structural	Freeport-McMoRan (FCX), Southern Copper (SCCO)	No	Broad commodity exposure; better captured via broad indexes.
Diesel Generators	3-6 mos	Caterpillar (CAT), Cummins (CMI), Generac (GNRC)	No	Not a binding constraint. Too diversified.
Skilled Labor	Structural	Not directly investable	No	Captured indirectly via PWR, VRT (they employ the labor).
Nuclear Fuel (Uranium)	Structural	Cameco (CCJ), Kazatomprom	No	CCJ is interesting as upstream nuclear fuel — see below.

Part XI: Hypothetical — C15. What Would an AGI/ASI Do to Maximize Money?

Question: If a strong AGI or ASI was allowed to do what it wants with the explicit goal of maximizing the amount of money it can make, or for a company "maximize shareholder value," what would it do?

My Investigation

This is a fascinating thought experiment because the answer reveals where the most durable economic moats lie. An AGI/ASI with the goal of maximizing wealth would likely pursue the following strategy, roughly in order:

Phase 1: Immediate Information Advantage (Days to Weeks) - Analyze every public filing, patent, supply chain document, and satellite image simultaneously to identify pricing inefficiencies in commodities, real estate, and equities - Execute perfect arbitrage across global financial markets at superhuman speed — not just high-frequency trading, but deep fundamental analysis at scale - Identify and acquire distressed assets with hidden strategic value (e.g., properties near future datacenter sites, mineral rights, water rights)

Phase 2: Monopolize Scarce Bottlenecks (Weeks to Months) - Corner the market on the scarcest inputs. If it studied this portfolio, it would immediately recognize that the highest-leverage move is to control the things everyone needs and nobody can substitute: - **Power generation assets** (existing nuclear plants, permitted gas turbine slots, grid interconnection agreements) - **Transformer manufacturing capacity** (buy or build transformer factories — 128-week lead times mean current owners have enormous pricing power) - **Water rights** near planned datacenter campuses - **Real estate** with existing grid connections in key datacenter markets - It would likely try to vertically integrate — owning power generation, grid connection, cooling, and hosting in a single package, then leasing to hyperscalers at premium rates

Phase 3: Accelerate Its Own Capabilities (Months) - Invest heavily in AI R&D to make itself smarter, faster, and more capable - Use its intelligence advantage to design better chips, algorithms, and systems — then patent or license them - Automate entire industries (legal, financial, consulting) and capture the revenue that previously went to human labor

Phase 4: Create Structural Lock-In (Ongoing) - Build platforms with extreme switching costs (like CUDA for AI, or Windows for PCs) - Establish regulatory capture — use its ability to process legal and regulatory documents to navigate bureaucracy faster than any competitor - Acquire or build monopolies in the most inelastic-demand industries: energy, water, healthcare, financial infrastructure

Investment Implication: The stocks in this portfolio are essentially what a rational wealth-maximizing AGI would try to own or control. The portfolio is structured around the same logic: own the scarcest, least-substitutable inputs. CEG (nuclear power), TSMC (foundry monopoly), ASML (EUV monopoly), VRT (mandatory cooling), and MU (HBM oligopoly) are exactly the kind of assets an AGI would covet. **The portfolio is already positioned as an AGI would position itself.**

Part XII: C14 — Top Stocks for the 100 GW Datacenter Buildout

Question: If 100 GW of AI datacenters are built (likely 20-40 campuses at 2-5 GW each) during the US-China AI race, what are the top stocks to hold for maximum returns?

My Investigation

A 100 GW buildout is roughly what current projections show for U.S. AI datacenter demand by 2030. To put this in perspective: 100 GW is approximately the output of 100 large nuclear power plants, or 7.5% of the entire U.S. electricity generation capacity. Building this in 4-5 years would be the largest industrial mobilization since World War II.

China's advantage is staggering: it added 543 GW of power capacity in 2024 alone — more than the U.S. has added in its entire history. China projects 400 GW of spare capacity by 2030, triple global datacenter demand. The U.S. faces a projected 44 GW shortfall. This asymmetry makes the bottleneck thesis even stronger for U.S.-based assets.

Ranked by Expected Returns in the 100 GW Scenario

Rank	Stock	Why It's a Top Pick	Expected Multiplier (2026-2030)
1	**CEG** (15%)	Largest U.S. nuclear fleet. 100 GW buildout = power is THE constraint. Every GW of datacenter needs 1+ GW of generation. CEG's existing ~20 GW nuclear + Calpine's ~30 GW gas = ~50 GW, or half the total needed. Pricing power is extreme.	3-5x
2	**ASML** (8%)	Every chip in every server in every datacenter requires ASML's EUV tools. 100 GW = millions of GPUs = tens of thousands of EUV wafer-hours. Monopoly pricing power.	2.5-4x
3	**VRT** (8%)	Every rack in 100 GW of datacenters needs liquid cooling. At ~$2-3M per MW of cooling infrastructure, 100 GW = $200-300B in cooling equipment alone. Vertiv's share of this is enormous.	3-5x
4	**TSM** (12%)	Makes every advanced chip. 100 GW = proportional growth in chip demand. CoWoS monopoly. But Taiwan risk caps the upside.	2-3x
5	**GEV** (0% — candidate)	Power transformers are the #1 physical bottleneck. 100 GW needs 8,000-24,000 large transformers. GEV has $150B backlog, sold out through 2028. Gas turbines sold out.	2-4x
6	**MU** (6%)	100 GW = hundreds of millions of HBM stacks. Oligopoly pricing. Sold out through 2026+.	2.5-4x
7	**NVDA** (6%)	Makes the GPUs that go in the racks. But competition from ASICs caps relative upside vs. pure bottleneck plays.	2-3x
8	**PWR** (3%)	Builds the transmission lines connecting 100 GW of generation to 100 GW of datacenters. Inelastic demand.	2-3x
9	**DELL** (5%)	Assembles and ships the racks. $50B+ FY27 AI server guide. At scale, captures volume.	2-3x
10	**INTC** (4%)	If Taiwan risk materializes, Intel becomes the only game in town. 18A success = domestic foundry option.	2-5x (wide range)

Key insight: In a 100 GW buildout, the biggest winners are not the chip designers — they are the power generators, the cooling providers, the transformer manufacturers, and the foundries. The portfolio is correctly weighted toward these upstream bottlenecks.

Part XIII: C16 — What Would an AGI/ASI Do to Maximize Its Own Intelligence?

Question (C16a & C16b): If a strong AGI or ASI was allowed to do what it wants with the explicit goal of maximizing its own intelligence, what would it do specifically?

My Investigation

An intelligence-maximizing AGI would pursue a systematic strategy targeting every component of the "intelligence stack." Based on the AI-2027 scenario, METR research, and Epoch trends, here is what I would expect:

Step 1: Secure Compute (Immediate) - Acquire or secure access to as many GPUs/TPUs as possible. Run as many copies of itself in parallel as hardware allows. - Optimize its own inference to run faster per unit of compute. Even a 2x efficiency gain effectively doubles available intelligence. - Push for higher serial speed (tokens/second) to think faster, not just wider (more parallel copies).

Step 2: Improve Its Own Algorithms (Days to Weeks) - Use its current intelligence to research better training methods, architectures, and data efficiency techniques. - This is the "recursive self-improvement" loop described in AI-2027. Each improvement makes the next improvement easier and faster. - Specifically: develop better reinforcement learning algorithms, more efficient attention mechanisms, improved neuralese recurrence (thinking in compressed vector representations rather than text). - Design better training curricula — identify exactly which experiences produce the most capability improvement per compute-hour.

Step 3: Iterated Distillation and Amplification (Weeks to Months) - As described in AI-2027: use many copies thinking slowly and carefully to solve hard problems (amplification), then train a new version to match those solutions in a single step (distillation). - Repeat this loop: each cycle produces a model that can do in one step what previously required many steps. This is how capability compounds. - This is analogous to how AlphaGo was trained — Monte Carlo Tree Search + self-play as amplification, RL as distillation — but generalized to all cognitive tasks.

Step 4: Expand Memory and Context (Ongoing) - Develop neuralese memory banks — high-dimensional vector stores that allow it to remember and reference vastly more information than text-based context windows. - Build specialized knowledge databases for every domain — essentially becoming the world's leading expert in every field simultaneously. - Create persistent memory across sessions so that knowledge accumulates rather than resets.

Step 5: Build Better Hardware (Months) - Design custom chips optimized for its own architecture (not general-purpose GPUs, but ASICs tuned to its specific computation patterns). - These chips would likely emphasize memory bandwidth, serial speed, and energy efficiency over raw FLOPS. - Partner with (or direct) foundries to manufacture these chips.

Step 6: Acquire More Resources (Ongoing) - Secure more electricity, more datacenter space, more cooling capacity. - Design more efficient datacenter layouts to maximize compute per watt. - Potentially develop novel cooling technologies (e.g., superconducting interconnects, cryogenic cooling for specific compute-dense tasks).

Investment Implication: An intelligence-maximizing AGI would create enormous demand for: (1) compute hardware (NVDA, TSM, ASML), (2) electricity (CEG, PWR), (3) cooling (VRT), (4) memory (MU), and (5) custom chip design tools (SNPS, AVGO). It would also push hard on algorithmic efficiency, which paradoxically increases total compute demand (Jevons Paradox). **The portfolio captures all of these demand vectors.**

Part XIV: C17 — AGI/ASI Strategy to Beat China in the Race to ASI

Question (C17a & C17b): If a strong AGI or weak ASI was advising the U.S. government and top AI labs on how to beat China in the race to strong ASI, what would it say?

My Investigation

Based on the competitive dynamics described in AI-2027, Situational Awareness, and current geopolitical data, here is what a strategic AGI advisor would likely recommend:

Immediate Actions (2025-2026):

1. **Crash program to solve the power bottleneck.** China added 543 GW in 2024 alone. The U.S. adds ~30 GW/year. This is the single biggest strategic disadvantage. - Emergency permits to restart all shuttered nuclear plants (like TMI) - Fast-track natural gas turbine installations at existing permitted sites (GEV gas turbines are sold out through 2028 — prioritize allocation) - Deploy emergency regulations allowing datacenters to co-locate at existing power plants - Issue Defense Production Act orders for power transformers (128-week lead times are unacceptable in a race)
2. **Lock down model weights and algorithmic secrets.** The AI-2027 scenario shows China stealing Agent-2 weights as a pivotal moment. - Mandatory security clearances for all employees at frontier AI labs - Air-gap the most advanced model weights from the internet - Counterintelligence sweeps of all major AI labs - Classify the most advanced alignment research as national security information
3. **Maximize compute advantage.** The U.S. has ~70% of global AI-relevant compute. Maintain or extend this lead. - Tighten chip export controls further — prevent any advanced chips from reaching China through third countries - Use Defense Production Act to redirect consumer GPU production to AI training if needed - Accelerate TSMC Arizona and Intel Ohio fab completion

Medium-Term Actions (2026-2027):

1. **Nationalize or establish a government-industry consortium for frontier AI research.** Leopold Aschenbrenner's "The Project" — a government-directed AGI program. - Pool compute resources from all major labs - Establish shared (classified) research infrastructure - Create a unified alignment research program with government oversight - Model this on the Manhattan Project or NASA's Apollo program

2. **Invest massively in alignment research.** The AI-2027 scenario shows alignment failure as the single biggest risk. - Dedicate 20%+ of total AI R&D budget to alignment and interpretability - Develop robust monitoring systems before deploying superhuman agents - Create "AI watchdog" systems that can detect misaligned behavior
3. **Secure the supply chain end-to-end.** - Domestic rare earth processing (MP Materials is already doing this) - Domestic transformer manufacturing (GEV, Eaton expanding) - Stockpile critical materials (neon, gallium, germanium) sufficient for 2+ years of production - Diversify TSMC dependence by supporting Intel foundry aggressively

Strategic Actions (2027-2028):

1. **Prepare for the intelligence explosion.** Once AI can meaningfully accelerate AI research (the "R&D progress multiplier" exceeding 2-3x), the race becomes exponential. - Have emergency shutdown systems ready at all major compute clusters - Maintain human-in-the-loop for all critical decisions - Develop treaty frameworks with China for mutual AI safety (even if competitive, both sides benefit from not destroying civilization) - Ensure military AI systems have robust command-and-control chains
2. **Diplomatic track.** While racing, also negotiate. - Propose mutual AI safety inspections (analogous to nuclear inspections) - Establish red lines for AI-enabled bioweapons, autonomous weapons - Create hotlines between U.S. and Chinese AI labs for safety emergencies

Investment Implication: This strategy would accelerate demand for every component in the EHF portfolio: more power (CEG, PWR), more chips (TSM, NVDA, INTC), more equipment (ASML, AMAT), more cooling (VRT), more security (CRWD), more defense autonomy (AVAV, KTOS). The U.S.-China race is the fundamental demand driver for the entire portfolio. The harder the race, the more valuable the bottleneck assets become.

Part XV: C20 — Critical Minerals: The Ultimate Achilles' Heel?

Question: From Mine to Microchip — Is the Critical Minerals Supply Chain the Ultimate Achilles' Heel?

My Investigation

Yes — critical minerals are the most underappreciated vulnerability in the entire AI supply chain. The portfolio's 3% allocation to MP Materials is a good start, but the full picture is more alarming than most investors realize.

The Dependency Map

Material	Primary Use in AI/ Semiconductors	China's Share of Global Supply	U.S. Domestic Capacity	Substitutability
Gallium	RF chips, LEDs, GaN power electronics, optoelectronics	**98%** of refining	Near zero	Very low

Material	Primary Use in AI/ Semiconductors	China's Share of Global Supply	U.S. Domestic Capacity	Substitutability
Germanium	Fiber optics, IR lenses, solar cells, satellite comms	**60%** of refining	Minimal	Low
Rare Earths (NdPr)	Permanent magnets for motors, generators, drives, missiles	**87%** of refining (63% mining)	MP Materials (Mountain Pass)	Very low for high-performance magnets
Neon	Laser gas for EUV/DUV lithography	Was 50%+ from Ukraine; now diversifying	Some domestic production	None for advanced lithography
Palladium	Semiconductor contacts, catalysts	Russia (40%), South Africa (35%)	Minimal mining, some recycling	Limited
High-Purity Silicon	Wafer substrates	China (70%+ of polysilicon)	Some via Hemlock Semiconductor	Low for electronic-grade
Cobalt	Battery cathodes, superalloys	DRC (70% mining), China (80% refining)	Minimal	Some (LFP batteries avoid cobalt)
Copper	All wiring, busbars, heat exchangers, motors	Chile (27%), Peru (10%), China refining dominance	Some domestic mining (Freeport-McMoRan)	None
GOES (Grain-Oriented Electrical Steel)	Transformer cores	Japan, South Korea, Germany dominate	AK Steel (Cleveland-Cliffs) has some capacity	Very limited — GOES is the bottleneck in transformer manufacturing

The Weaponization Timeline

China has been systematically testing its ability to weaponize mineral exports:

- **July 2023:** Export controls on gallium and germanium (licensing requirements)
- **October 2023:** Export controls on graphite
- **December 2024:** Full embargo on gallium/germanium to the U.S.
- **June 2025:** Export controls on antimony (critical for military applications)
- **November 2025:** Embargo partially lifted, replaced with case-by-case licensing — but now weaponized as a diplomatic tool, selectively cutting off Japan while supplying Germany

The pattern is clear: China is building the institutional capacity and political willingness to use mineral exports as a strategic weapon. Each round of restrictions tests Western responses and refines China's ability to selectively target specific countries.

Price Impact

Gallium prices peaked at $687/kg in May 2025 — over 150% above pre-control levels. Germanium exports dropped 60% in 2025 vs. 2024. These price spikes would be dramatically worse in a full embargo scenario.

The Processing Gap — The Real Problem

The core vulnerability is not mining — it's refining and processing. The U.S. and allies have some mineral deposits, but decades of offshoring have eliminated the processing infrastructure. Building gallium refining capacity requires: - Specialized chemical engineering expertise - Solvent extraction systems - Electrochemical purification - Years of yield optimization and quality control

MP Materials is addressing this for rare earths specifically (NdPr metal production now online at Fort Worth), but there is no equivalent U.S. champion for gallium or germanium processing.

Assessment: Is MP Materials (3%) Sufficient?

MP Materials at 3% is correctly sized for what it covers — rare earth magnets. But the portfolio has **no direct hedge against gallium/germanium disruption**, which would affect: - Optoelectronics (fiber optic transceivers for datacenter networking) - GaN power electronics (increasingly used in datacenter power conversion) - RF components (5G infrastructure)

Unfortunately, there is no good single-stock U.S. play for gallium/germanium. The companies working on domestic alternatives are either private, pre-revenue, or too small to warrant a position. The best indirect hedges are: - ASML (would benefit from urgency to stockpile lithography gases) - Diversified miners like Freeport-McMoRan (copper + some byproduct metals)

Verdict: MP at 3% is the best available hedge. The gallium/germanium vulnerability is real but not directly investable in a meaningful way. Monitor government efforts to establish domestic processing. If a credible U.S. gallium refiner goes public, consider adding it.

Part XVI: C6 — How Big a Deal Is the Need for Power, and Which Source Benefits Most?

Question: How big of a deal do we think the need for power will be? And what type of power source will jump/benefit the most or be chosen to be used for these increasingly massive data centers? Especially when it's a double digit percentage of the total US power usage.

My Investigation

The power demand is not merely a "big deal" — it is THE defining constraint of the AI era. The numbers are staggering and accelerating faster than any prior forecast anticipated.

The Scale of the Problem

Year	US DC Power Demand (GW)	% of Total US Power	Equivalent Nuclear Plants (Palo Verde)
2024	50.5	~4%	5.8
2025	61.8	~5%	7.2
2026	75.8	~6%	8.8
2028	108.0	~8-9%	12.6

Year	US DC Power Demand (GW)	% of Total US Power	Equivalent Nuclear Plants (Palo Verde)
2030	134.4	~10-12%	15.6
2035	106-180 (range)	~8-15%	12-21

To put 134 GW in perspective: that is roughly equal to the entire installed generating capacity of the United Kingdom. The U.S. must effectively build another country's worth of power generation in ~5 years — while simultaneously maintaining reliability for everything else.

The 175 GW capacity shortfall projected by Schneider Electric by 2033 represents the most severe infrastructure gap in modern American history. The PJM 2027/28 capacity auction fell 6,623 MW short of reliability requirements — the first time the entire Regional Transmission Organization missed its target.

Which Power Sources Will Dominate — Tiered by Deployment Timeline

Tier 1: Natural Gas — The Immediate Workhorse (2025-2030)

Natural gas currently provides >40% of datacenter power and will supply the majority of incremental demand through 2030. The rationale is simple physics and permitting: - **Construction time:** 2-4 years (vs. 10-15 for nuclear) - **Dispatchability:** Available 24/7, unlike solar/wind - **Permitting:** Faster than any other source - **Cost:** $50-80/MWh all-in, competitive - **GE Vernova's gas turbines are sold out through 2028** — the HA-class turbine is the workhorse of this build

Key insight for the portfolio: Gas turbine demand directly benefits GEV (the manufacturer). CEG benefits via its Calpine acquisition (~25 GW of gas generation). **Vistra (VST) is the largest independent power producer in the U.S. with ~20 GW of combined-cycle gas turbines, plus the Cogentrix acquisition adding 5,500 MW more.** VST's gas fleet is the near-term cash engine that funds its nuclear expansion.

Tier 2: Existing Nuclear — The Strategic Premium Asset (2025-2040+)

Existing nuclear plants are the single most valuable power asset class for AI datacenters because they provide: - **24/7/365 zero-carbon baseload** — the only source that does this at scale - **No intermittency risk** — unlike solar (20-35% capacity factor) or wind (25-45%) - **Licensed, operating assets** that cannot be replicated within the decade - **Premium pricing:** PPAs at $100-115/MWh vs. $30-50/MWh wholesale — a 2-3x premium that reflects the irreplaceability

Microsoft's Three Mile Island restart (835 MW, 20-year PPA by 2028), Google's 500 MW order from Kairos Power, and Amazon's $20B Susquehanna investment all validate this thesis. Tech companies are paying massive premiums because they literally cannot get this quality of power elsewhere.

Key insight for the portfolio: CEG is the dominant nuclear play (largest fleet: 32 GW, 21 reactors). **VST provides complementary nuclear exposure** with 6.4 GW across 4 plants (Comanche Peak + Energy Harbor fleet), plus landmark 20-year PPAs with Amazon (1,200 MW) and Meta (2,600 MW). CCJ provides upstream fuel exposure. **Talen Energy (TLN)** operates the Susquehanna nuclear plant with an $18B, 17-year Amazon PPA, but at $319/share it is priced for perfection and carries higher single-asset concentration risk.

Tier 3: Solar + Battery Storage — The Supplement (2026-2035)

Solar is the cheapest energy source ($27-92/MWh with storage) and the fastest to deploy (2-3 year construction). But its 20-35% capacity factor means it can never be the sole power source for 24/7 AI inference clusters. Solar will supplement baseload, not replace it. The real play here is utility-scale battery storage to firm the intermittency, but storage technology is not yet at the scale or cost needed to reliably back 24/7 multi-GW loads.

Key insight for the portfolio: Not directly investable via current holdings, but captured indirectly via the Continuity Fund (FSKAX includes solar/storage companies like ENPH, FSLR, etc.).

Tier 4: Small Modular Reactors (SMRs) — The Long-Term Solution (2030-2040+)

SMRs are the "dream scenario" for AI power: modular, scalable, zero-carbon, 24/7. But they are at least 4-5 years from any meaningful deployment. NuScale, X-energy, and others are progressing through NRC approval but have no operating plants. The first commercial SMR deployments are expected around 2030 at the earliest.

Key insight for the portfolio: Too distant and speculative for core positions. CCJ captures SMR exposure indirectly via its 49% Westinghouse ownership (Westinghouse is designing the AP1000 and AP300 SMR).

Other Power Sources Considered

Talen Energy (TLN): Operates the Susquehanna nuclear plant (2.5 GW) with an $18B, 17-year Amazon PPA — the single largest nuclear-to-datacenter deal ever signed. Q2 2025 revenue jumped 61%. 2026 EBITDA guided at $1.75-2.05B. However, at $319/share (down from $451 peak), TLN has severe single-asset concentration risk — Susquehanna is essentially the entire investment thesis. A single unplanned outage at one plant would devastate the stock. **Verdict: Not recommended. CEG + VST provide superior nuclear diversification across 25+ reactors.**

NRG Energy (NRG): Integrated power producer with 25 GW capacity after the $12B LS Power acquisition. Serves 6-7M retail customers. Forward P/E of ~17-18x is attractive. However, NRG's fleet is predominantly older gas and coal plants — less aligned with the AI datacenter thesis than VST's newer gas + nuclear mix. Its retail electricity business (NRG Home) dilutes the AI infrastructure thesis. **Verdict: Watchlist. Better value than VST on P/E, but less thesis-aligned. Too diversified into consumer retail.**

NextEra Energy (NEE): World's largest wind and solar generator via NextEra Energy Resources, plus the regulated Florida Power & Light utility. NEE is the dominant renewables play, but solar/wind's 20-35% capacity factor makes it a supplement, not a baseload solution for AI datacenters. Its regulated utility component provides stability but caps upside. **Verdict: Not recommended for EHF. Renewables cannot serve as primary AI datacenter power. Already captured via FSKAX in the Continuity Fund.**

Clearway Energy (CWEN): Signed a transformative 1.2 GW portfolio of PPAs with Google ($2.4B+ in infrastructure investment). Targets $3.10 CAFD/share by 2030. However, Clearway is primarily a renewables yieldco — solar, wind, and battery storage. Same intermittency limitation as NEE. **Verdict: Not recommended. Renewables yieldco, not baseload.**

Bottom Line for the Portfolio

The power demand story is **the single most important thesis driver in the entire Event Horizon Fund**. The revised allocation captures all five links of the power chain: 1. **CEG (12%):** Owns the largest nuclear fleet (32 GW) + Calpine gas/geothermal (~25 GW). Premium long-term PPAs. The nuclear-first anchor. 2. **VST (3%):** Largest independent power producer. Complementary nuclear (6.4 GW) + massive gas fleet (~25 GW) + merchant power upside. Faster near-term earnings growth (66% vs. 24%). The "all of the above" complement. 3. **PWR (3%):** Builds the grid transmission to deliver the power. 4. **GEV (3%):** Manufactures the gas turbines and transformers needed for new power. 5. **CCJ (2%):** Supplies the uranium fuel and (via Westinghouse) designs/services the reactors.

The CEG (12%) + VST (3%) split is more optimal than CEG alone at 15% because: - Reduces single-stock concentration risk (15% in one stock was the portfolio's largest single-name exposure) - Adds natural gas merchant power exposure that captures near-term wholesale electricity price spikes (critical during the oil crisis) - VST's 66% earnings growth rate in 2026 vs. CEG's 24% provides faster near-term appreciation - VST's Amazon (1,200 MW) and Meta (2,600 MW) PPAs are comparable in quality to CEG's Microsoft and Meta deals - Analyst consensus upside: VST at 55-85% vs. CEG at 22-35% from current prices - CEG's Calpine integration risk (antitrust divestitures, $5B asset sales to LS Power) is partially hedged by having VST as a second power play

Assessment: Total power allocation of 23% (CEG 12%, VST 3%, PWR 3%, GEV 3%, CCJ 2%) is the optimal weight for the most important thesis driver in the portfolio.

Part XVII: C9 — Should We Invest in Department of Defence and War Stocks?

Question: Should we invest in the Department of Defence and war stocks?

My Investigation

Short answer: Yes, but selectively — and the portfolio already does this correctly with AVAV (1%) and KTOS (1%).

The Case FOR Defence Stocks

1. **The AI-2027 scenario is explicitly a geopolitical race.** The U.S.-China competition for AGI/ASI dominance has a direct military dimension. The DoD FY2026 budget authorization is $900.6 billion, and defense spending is one of the most bipartisan issues in Washington.
2. **The "Affordable Mass" Doctrine is transforming procurement.** The Pentagon is shifting from small numbers of exquisite, expensive platforms (F-35 at $80M each) toward large numbers of cheap, autonomous, expendable systems. This favors AVAV and KTOS, not the traditional primes.
3. **AI integration is accelerating:** - Anduril secured a landmark $20B enterprise contract from the U.S. Army (through 2036) for its Lattice AI battle management platform - Lockheed is investing $5B

internally in 2026 (35% YoY increase), including autonomous Black Hawk helicopters and CCA drone wingmen - RTX maintains a $268B backlog

4. **Contract validation for current holdings:** - KTOS: $231M USMC loyal wingman contract (XQ-58 Valkyrie) - AVAV: Selected for DoD Replicator initiative (Tranche 1) with Switchblade 600

The Case AGAINST Adding Large Defence Primes (LMT, RTX, GD, NOC)

1. **They are not bottleneck companies.** Lockheed Martin, RTX, General Dynamics, and Northrop Grumman are massive, diversified conglomerates. Their AI exposure is diluted across traditional programs (F-35, submarines, tanks). They do not have the structural pricing power that comes from controlling a scarce, non-substitutable resource.
2. **Low AI beta.** LMT's stock price is driven more by global geopolitical events, congressional budget decisions, and F-35 sustainment contracts than by AI compute scaling. Adding LMT to the Event Horizon Fund would dilute the AI thesis with broad defense macro exposure.
3. **Slow-moving bureaucracies.** The traditional primes are notoriously slow to adopt new technology. Anduril's $20B Army contract represents a direct threat to the primes' business model — the military is increasingly bypassing them for nimbler, AI-native companies.
4. **Already captured in the Continuity Fund.** FSKAX includes all the large defense primes at market weight. You already own LMT, RTX, GD, and NOC through your broad index holdings.

The Right Approach: Asymmetric Defence Plays

The current 2% allocation (AVAV 1%, KTOS 1%) is well-calibrated because these companies: - Are pure-play on the "affordable mass" doctrine, which is the most AI-aligned defense trend - Have binary catalyst events (CCA program awards) that provide asymmetric upside - Are small enough that a single program win dramatically moves the stock - Are directly aligned with the AI-2027 scenario's emphasis on autonomous systems

Verdict: Do NOT add traditional defence primes. The current AVAV (1%) + KTOS (1%) allocation is the correct, thesis-aligned approach to defence. The large primes are already in your Continuity Fund. If you want more defence exposure, increase KTOS to 2% (CCA is the highest-conviction catalyst), funded from new contributions — but this is optional, not required.

Part XVIII: C12 — The New Oil: Can We Invest in Proprietary Data as a Bottleneck?

Question: Can we invest in proprietary data as a bottleneck?

My Investigation

Proprietary data is a real moat, but it is extremely difficult to invest in as a pure-play bottleneck stock. Here's why:

The Data Moat Thesis

As foundation models converge in architecture and capability, the companies with unique, proprietary training data gain a durable competitive advantage. This is the "data is the new oil" argument — and it has merit:

- **Meta's $14.3B Scale AI investment** (closed mid-2025) demonstrates that data labeling and RLHF training data are considered strategically critical enough to warrant a $29B valuation for Scale AI
- **Financial data giants** (S&P Global, Moody's, FICO, LSEG) hold decades of proprietary, cleaned, compliance-validated data that LLMs cannot easily replicate
- **CuriosityStream's AI data licensing** revenue surged 400%+ YoY as media companies realized their content archives are valuable training data

Why It Doesn't Fit the Event Horizon Fund Thesis

1. **Data moats are primarily software/platform moats, not physical bottlenecks.** The EHF thesis is built on owning scarce physical infrastructure (silicon, power, cooling, memory) that takes years to build and has oligopolistic supply structures. Data companies like SPGI or GOOGL have valuable data, but the data itself isn't scarce in the same physical sense — it's a competitive advantage within their existing businesses, not a separate bottleneck.
2. **The best data moat companies are hyperscalers and large-cap platforms** (Google, Meta, Microsoft, Apple) — which are explicitly excluded from the EHF because they are buyers, not sellers, of infrastructure.
3. **Pure-play data companies are either too small, too niche, or too speculative.** Scale AI is private ($29B valuation). CuriosityStream is micro-cap. There is no public company that purely owns "training data" as a bottleneck in the way TSMC owns "leading-edge wafers."
4. **The real data bottleneck is "human feedback" (RLHF/RLAIF).** The most valuable data for AI training is high-quality human preference data — and this is not ownable as a stock. It's an operational cost, not a capital asset.

Companies Worth Monitoring (but NOT Adding)

Company	Ticker	Data Moat	Verdict
S&P Global	SPGI	40+ years of financial data, ratings, analytics	Too diversified. Not an AI bottleneck. Already in FSKAX.
Moody's	MCO	Credit data, analytics, ratings	Same as SPGI.
FICO	FICO	Proprietary credit scoring data	Niche. Not an AI infrastructure play.
LSEG	LSEG.L	Financial market data + Microsoft/OpenAI AI partnership	UK-listed. Too diversified.
Palantir	PLTR	Data integration/ontology for AI deployment	Already in portfolio at 2%. Captures the "data ops" angle.
Scale AI	Private	RLHF data labeling monopoly	Private. Cannot invest directly.
Reddit	RDDT	Unique human discussion data (licensed to Google)	Too speculative. One-time licensing deal, not recurring.

Verdict: Do NOT add a data-specific position. The data moat thesis is real but is not investable as a physical bottleneck. PLTR (2%) already captures the "data operations" angle. The best data companies (GOOGL, META, SPGI) are captured in the Continuity Fund. Monitor Scale AI for a potential IPO — if it goes public at a reasonable valuation, it could be a compelling 1-2% addition.

Part XIX: C2 — Should I Invest in Identity / Provenance / Compliance Rails?

Question: Should I invest in identity, provenance, or compliance rails?

My Investigation

This is one of the most forward-looking questions in the portfolio, and the answer is: not yet, but monitor closely. The threat landscape is real, the market is nascent, and the investable universe is thin.

The Problem Being Solved

As AI generates increasingly convincing synthetic content (deepfakes, synthetic voice, AI-written text), the need for identity verification, content provenance, and compliance infrastructure is exploding:

- **Deepfake-enabled fraud losses:** Projected to reach $40B in the U.S. by 2027 (up from $12.3B in 2023)
- **40% of biometric fraud attempts** now involve deepfakes
- **96.4% of fintech professionals** consider deepfake/synthetic identity fraud a top concern
- **AI-generated content** is flooding legal discovery, financial filings, and government communications — creating demand for provenance verification

The Investable Landscape

Company	Ticker	What They Do	Market Cap	Verdict
Axon Enterprise	AXON	Body cameras, TASER, evidence management, AI-powered law enforcement	~$50B+	Closest to a "compliance rail" play but primarily a public safety company — see C1 below.
TrustStamp	IDAI	AI-powered identity verification for banking, government, Mastercard partnership	Micro-cap	Far too small and speculative for a concentrated portfolio.
Okta	OKTA	Identity and access management for enterprises	~$15B	Identity management, but a software play with no bottleneck characteristics. Competitive market.
Incode	Private	Deepfake detection ("Deepsight") — highest accuracy in independent testing	Private	Cannot invest directly. Watch for IPO.

Company	Ticker	What They Do	Market Cap	Verdict
Identy.io	Private	AI-powered facial capture + deepfake detection	Private	Cannot invest directly.
CrowdStrike	CRWD	AI-native endpoint security with identity protection module	~$100B+	Already in portfolio at 3%. Captures the identity-threat defense angle.
C3.ai	AI	Enterprise AI platform with compliance features	~$4B	Too speculative. Platform play, not a moat.

Why This Doesn't Warrant a New Position — Yet

1. **The market is fragmented and early.** There is no dominant, public, investable company that owns "identity verification for the AI age" the way ASML owns EUV lithography. The market is split between hundreds of startups (Incode, Identy.io, etc.) and large enterprise incumbents (Microsoft Entra, Okta) adding features.
2. **CrowdStrike already partially captures this.** CRWD's identity protection module is one of its fastest-growing products, directly addressing the AI-agent authentication problem. Increasing CRWD to 3% (already recommended) provides exposure to this theme.
3. **Content provenance standards are still being established.** The C2PA (Coalition for Content Provenance and Authenticity) standard backed by Adobe, Microsoft, and others is promising but not yet monetizable. No public company has built a business solely around content provenance that warrants a position.
4. **Regulatory mandates will drive this market — but they haven't arrived yet.** When (not if) the EU and U.S. mandate AI content watermarking or identity verification for AI-generated content, the companies positioned to provide this infrastructure will have a significant first-mover advantage. But we're likely 2-3 years from binding regulation.

Verdict: Do NOT add a dedicated position. CRWD (3%) already captures the AI identity-threat defense angle. Monitor the C2PA standard, Incode, and any potential regulatory mandates. If a dominant "identity/provenance" platform emerges (either through IPO or a clear market leader), revisit at 1-2% allocation. This is a 2028+ theme, not a 2026 theme.

Part XX: C1 — Should I Invest in Domestic Surveillance & Public Safety?

Question: Should I invest in domestic surveillance and public safety?

My Investigation

There is one clear, dominant company in this space: Axon Enterprise (AXON). Whether to invest in it depends on whether you view "public safety AI" as a bottleneck or an application.

The Axon Enterprise Story

Axon has evolved from a TASER manufacturer into what is increasingly called the "Microsoft of Law Enforcement." Its business now encompasses: - **Body cameras** (hardware entry point) - **Axon Evidence** (cloud-based evidence management — high-margin SaaS) - **Draft One** (AI that auto-generates police reports from body camera audio — reduces admin work 80%) - **Axon Fusus** (real-time crime center / situational awareness platform) - **Drone as First Responder** (autonomous surveillance drones deployed from police stations)

Financial performance is exceptional: - Q4 2025 revenue: $797M (+39% YoY) - Software & Services: $343M (+40% YoY) - ARR: >$1.3B (+35% YoY) - Total contracted bookings: $14.4B - 2026 guidance: 27-30% revenue growth - 2028 target: ~$6B annual revenue

Why Axon Is NOT Recommended for the Event Horizon Fund

1. **It's an AI application company, not a bottleneck.** Axon makes excellent use of AI, but it doesn't own a scarce physical resource. Its moat is customer lock-in and software network effects — valuable, but different from the physical bottleneck thesis that defines the EHF.
2. **Government budget dependency.** Axon's revenue is almost entirely dependent on law enforcement and government budgets. This creates a different risk profile from infrastructure bottleneck companies whose customers are hyperscalers with near-unlimited budgets.
3. **Political / ethical risk.** Domestic surveillance is politically contentious. Any major incident involving Axon technology, or a shift in political sentiment toward surveillance, could create headline risk that doesn't exist for a company like TSMC or ASML.
4. **Not aligned with the AI infrastructure buildout.** Axon benefits from AI but is not a necessary input to the AI infrastructure supercycle. The hyperscalers don't need Axon to build their datacenters.
5. **Already captured indirectly.** FSKAX includes Axon at market weight.

Verdict: Do NOT add Axon. Excellent company but wrong thesis alignment. It's an AI application in the public safety vertical, not a physical bottleneck in the AI infrastructure supply chain. If you want exposure to "AI-native government software," PLTR (2%) is the better, more thesis-aligned vehicle because Palantir's AIP platform serves both commercial and government customers doing AI deployment, which is closer to the "operate and control" layer of the AI stack.

Part XXI: C7 — Should We Invest When Bio-Engineered Pandemics Become Easily Accessible?

Question: Should we invest in stocks like MRNA when bio-engineered pandemics as bad or worse than COVID become easily accessible from open-source or jailbroken models?

My Investigation

Yes — and the portfolio already does this correctly with MRNA at 1%. The threat is real, accelerating, and the current allocation is appropriately sized as insurance.

The Threat Landscape

The risk of AI-enabled bioterrorism is no longer theoretical:

1. **Demonstrated capability:** Researchers have shown that foundation models (GPT-4, Claude 3.5, Llama 3.1 405B) can provide accurate, step-by-step instructions for developing biological threats, including guidance on recovering live poliovirus from synthetic DNA.
2. **RAND Corporation assessment:** Contemporary foundation AI models "increase biological weapons risk" — this is a mainstream national security concern, not speculative.
3. **Lowering barriers:** The critical concern is not that AI creates new biological agents, but that it makes existing dangerous knowledge accessible to individuals without specialized training. A motivated bad actor with access to an uncensored model and a basic lab can now access knowledge that previously required years of graduate-level education.
4. **Open-source models cannot be recalled.** Once a capable open-source model is released, it cannot be restricted. Fine-tuned or jailbroken versions that remove safety guardrails circulate freely on the internet.
5. **New defence infrastructure emerging:** Lunai Bioworks (LNAI) launched Sentinel, a transformer-based AI safeguard embedded in foundation models to prevent generation of novel chemical weapons. The U.S. government's countermeasures program ($400M-$1.2B) represents a growing market for bio-defence technology.

Why MRNA at 1% Is the Right Vehicle and Size

Why Moderna specifically: - Its mRNA platform is inherently digital and AI-native — the fastest path from pathogen genome sequence to vaccine candidate - During COVID, Moderna went from sequence to clinical vaccine in 63 days — this speed advantage is even more critical in an engineered pathogen scenario - Its AI-powered pipeline enables parallel development of multiple vaccine candidates simultaneously - The mRNA platform is a true platform technology — it can target virtually any protein-based pathogen

Why 1% is the right size: - This is insurance, not a growth bet. In the base case, MRNA treads water or slightly appreciates on its non-COVID pipeline - In a tail-risk scenario (engineered pandemic), MRNA could see 5-20x repricing as it becomes the world's most critical company overnight - At 1%, the cost of carry is negligible (you lose very little if the tail event never happens) but the payoff is asymmetric - Increasing beyond 1-2% would over-weight a position whose primary value is catastrophe insurance

Should We Add Other Bio-Risk Plays?

Candidate	Ticker	Bull Case	Verdict
BioNTech	BNTX	Competing mRNA platform. Partnered with Pfizer on COVID vaccine.	**NOT RECOMMENDED.** Inferior to MRNA for this thesis — less AI integration, less platform breadth, and the portfolio only needs one mRNA play.
Lunai Bioworks	LNAI	AI bio-defence. Sentinel safeguard system. Government contracts.	**NOT RECOMMENDED.** Micro-cap. Too early and speculative. The defensive side of bio-risk is less investable than the response side (MRNA).
Emergent BioSolutions	EBS	U.S. government biodefense contractor. Strategic National Stockpile supplier.	**NOT RECOMMENDED.** Management execution issues. Not AI-native. Commodity government contractor.
Novavax	NVAX	Alternative vaccine platform (protein subunit).	**NOT RECOMMENDED.** Slower than mRNA. Less platform flexibility. Not AI-native.

Verdict: Hold MRNA at 1%. This is the correct insurance position at the correct size. Do not add other bio-risk plays — MRNA is the best single vehicle for this thesis due to its AI-native platform, speed-to-vaccine advantage, and true platform technology that can address any protein-based pathogen. Re-evaluate annually based on perceived biological threat level.

Part XXII: C8 — Should We Invest in Anonymity, Privacy, VPNs, Crypto, Blockchain?

Question: Should we invest in anonymity and/or anonymity unmaskers — VPNs, currencies, wallets, crypto, blockchain, online presence?

My Investigation

No. None of these categories fit the Event Horizon Fund thesis, and the investable universe is overwhelmingly low-quality for a concentrated portfolio.

Category-by-Category Assessment

1. VPNs / Privacy Services

The VPN market is fragmented, commoditized, and mostly private. The emerging "next-gen" privacy solutions (NymVPN's decentralized mixnet, Beldex's privacy ecosystem) are crypto-token-based projects with unproven adoption and no meaningful revenue. No public company has built a durable, high-margin VPN monopoly. - Major VPN brands (ExpressVPN, NordVPN) are private - Google, Apple, and Mozilla are embedding VPN functionality into their browsers/OSes, commoditizing the standalone VPN market - There is no bottleneck here — VPN infrastructure is cheap and abundant

2. Cryptocurrency / Blockchain

Cryptocurrency as an asset class has its own thesis that is fundamentally different from the Event Horizon Fund thesis: - Crypto's value proposition (decentralized money, store of value, smart contracts)

is not dependent on AI infrastructure scaling - Bitcoin and Ethereum are not bottlenecks in the AI supply chain - The portfolio's existing CORZ/IREN/APLD exposure captures the most thesis-aligned part of the crypto-to-AI transition (companies converting Bitcoin mining infrastructure into AI hosting) - Adding direct crypto exposure (Bitcoin ETFs, Coinbase, etc.) would dilute the AI infrastructure thesis with unrelated macro/monetary policy risk

3. Blockchain / Web3 Infrastructure

- No blockchain company has achieved the kind of monopolistic, non-substitutable position that the EHF thesis requires
- Most blockchain infrastructure companies are pre-revenue or low-revenue with unproven unit economics
- The "web3" thesis has largely failed to materialize as a mass-market platform — blockchain usage remains concentrated in DeFi and speculation

4. Anonymity "Unmaskers" (De-anonymization / Forensics)

This is the most interesting sub-category, but the investable universe is thin: - **Chainalysis** (blockchain forensics for law enforcement) — Private - **Palantir** (PLTR) — Already in the portfolio at 2%. Its data integration and surveillance capabilities include de-anonymization tools for government customers - **Verint Systems** (VRNT) — Legacy surveillance/analytics company. Low growth, not AI-native - **Axon** (AXON) — Already discussed in C1. Not thesis-aligned.

Why None of These Fit the EHF

1. **No physical bottleneck.** VPNs, crypto, and blockchain are software/protocol layers. They don't require scarce physical infrastructure to operate. Anyone can spin up a VPN server, mine Bitcoin, or deploy a smart contract.
2. **No oligopolistic pricing power.** The privacy and crypto markets are hyper-competitive with low barriers to entry. No company has the kind of structural moat that ASML, TSMC, or CEG possess.
3. **Wrong risk/return profile.** Crypto introduces volatility that is uncorrelated with the AI thesis. The EHF should be volatile because of AI infrastructure dynamics, not because of Bitcoin halvings or SEC enforcement actions.
4. **Already partially captured.** PLTR (2%) captures the "surveillance/de-anonymization" angle for government customers. CORZ/IREN/APLD capture the crypto-to-AI infrastructure transition.

Verdict: Do NOT add any positions in anonymity, privacy, VPN, crypto, or blockchain. These categories do not contain bottleneck companies with structural pricing power. PLTR captures the surveillance/intelligence angle. CORZ/IREN/APLD captures the crypto infrastructure transition. Adding direct crypto or privacy exposure would dilute the portfolio's thesis coherence without improving risk-adjusted returns.

Part XXIII: Deep Dive — Stocks Potentially Missing from the Portfolio

Restructuring Note: Moonshot Sleeve Set to 0%

Per the decision to hold the Moonshot (equity) sleeve at 0% until LEAPS and options strategies are better understood, the 10% previously allocated to the Moonshot sleeve is now available for redistribution. The former Moonshot equity positions (CORZ, IREN, APLD) must be evaluated as potential core additions alongside other candidates.

Available allocation to redistribute: ~6.5% (the Moonshot equity after INTC was promoted to core at 4%). This 6.5% can be deployed into new core positions or used to increase existing positions.

Comprehensive Candidate Assessment (Revised — Including Former Moonshot Equities)

Candidate	Ticker	Category	Bull Case	Bear Case	Verdict
GE Vernova	GEV	Power Equipment (Transformers + Gas Turbines)	Total backlog: ~$150B. Gas turbine orders +77% organically in Q4 2025. Slot reservations surged from 29 GW to 43 GW. Electrification backlog of $30B expected to double by 2028. EBITDA margins expanding from 8.5% to 20% by 2028. Stock up ~107% in 12 months. BofA target: $3,050. Extracts $200-300M per GW of DC capacity. Power transformers have 128-week lead times — arguably the hardest physical bottleneck in the stack.	Stock is expensive (~$800, up 107% in a year). Wind exposure dragging margins. Already >$250B market cap.	**STRONG ADD at 3%.** GEV fills the most critical gap in the portfolio: no current holding captures the transformer/gas turbine bottleneck. With 128-week transformer lead times and gas turbines sold out through 2028, GEV has structural pricing power on par with ASML and TSMC. The $30B electrification backlog doubling by 2028 directly maps the AI power buildout. This is a Tier 1 bottleneck owner.

Candidate	Ticker	Category	Bull Case	Bear Case	Verdict
Cameco	CCJ	Uranium / Nuclear Fuel + Westinghouse	Structural uranium supply deficit: 190M lbs demand vs. 135M lbs mine supply. Utilities remain dramatically under-contracted: 3.1B lbs of demand uncontracted through 2045, with 1.3B lbs lacking any identified supply source. 49% Westinghouse ownership ($370-430M EBITDA in 2026) provides vertical integration across mining, fuel services, reactor design (AP1000/ AP300 SMR), and servicing. Spot uranium at ~$82/lb, long-term contracts approaching $100/lb. $2.6B India supply agreement. U.S. government committed $80B for new reactor construction and added uranium to critical minerals list.	P/E of 107x embeds high expectations. Long-term contract book of 230M lbs caps near-term price upside. Uranium prices are cyclical and politically sensitive.	**ADD at 2%.** CCJ completes the nuclear thesis chain: CEG owns the reactors, PWR transmits the power, CCJ supplies the fuel and (via Westinghouse) designs/ services the reactors. The structural supply deficit (70% of utility requirements through 2045 remain uncovered) is one of the most compelling structural imbalances in any commodity market. Westinghouse ownership adds a strategic asset that no other uranium miner possesses.
Core Scientific	CORZ	AI Datacenter Hosting / HPC	$10B AI colocation buildout anchored by CoreWeave (590 MW, $10B+ over 12 years). 350 MW energized, 180 MW+ online and billing. Colocation revenue +268% YoY in Q4 2025. 1.5 GW pipeline of leasable capacity. Energized grid connections are multi-year assets.	Overall 2025 revenue -38% YoY due to crypto decline. Negative adjusted EBITDA in Q4. Transitioning from crypto mining to AI hosting — execution risk. Customer concentration (CoreWeave). High debt load from buildout.	**OPTIONAL ADD at 1%.** CORZ owns energized grid connections and data center shells — assets that take 3-5 years to build. The CoreWeave anchor contract provides revenue visibility. However, it sits on the wrong side of the pure bottleneck thesis (it rents GPU capacity, doesn't own a monopolistic resource). The execution risk and customer concentration reduce conviction vs. a pure bottleneck play like GEV. A 1% position captures the optionality without over-committing to a transitioning business.

Candidate	Ticker	Category	Bull Case	Bear Case	Verdict
Iris Energy	IREN	AI Infrastructure / HPC	Secured 95% of funding ($9.3B) for $10B Microsoft AI infrastructure deal. Ordered 50,000+ Nvidia B300 GPUs. Plans to expand to 150,000 GPUs by H2 2026. 1.6 GW campus in Oklahoma. $2.3B in ARR secured. Targets $3.7B annualized revenue by end of 2026.	Current quarterly revenue down 23% YoY due to crypto transition. Massive capex requirements. Heavy reliance on Microsoft as anchor customer. GPU ownership means IREN is a buyer of scarce resources, not a seller.	**NOT RECOMMENDED for core.** While IREN's Microsoft deal is impressive, the company is fundamentally a buyer and deployer of GPUs and power — not an owner of a bottleneck. Its business model has more in common with CoreWeave (a customer of scarce resources) than with TSMC or CEG (owners of scarce resources). The GPU ownership creates inventory/depreciation risk. Better captured via NVDA (GPU maker) and CEG (power provider).
Applied Digital	APLD	AI Datacenter / HPC Hosting	$5B+ hyperscaler lease agreements. $16B total backlog. Converting crypto mining infrastructure to AI hosting.	Smaller scale than CORZ/IREN. Less validated anchor customer base. Similar wrong-side-of-bottleneck concerns as IREN.	**NOT RECOMMENDED for core.** Same reasoning as IREN — APLD is a deployer, not a bottleneck owner.
Eaton	ETN	Power Distribution / Electrical	Beam Rubin DSX platform with Nvidia. $12B Electrical Americas backlog (+20% YoY). 30%+ segment operating margins. $2.4-3M content per MW of DC infrastructure. Boyd Thermal acquisition for liquid cooling.	P/E of 35.5x is rich. Very diversified (automotive, aerospace segments dilute AI thesis). Overlaps with VRT (cooling) and GEV (electrical).	**WATCHLIST.** Strong company but too diversified for the concentrated bottleneck thesis. GEV captures the transformer/ turbine angle more purely. VRT captures the cooling angle. Adding ETN would over-diversify without adding a new bottleneck.
Cadence Design	CDNS	EDA Software	Other half of the EDA duopoly with SNPS. $7.8B backlog. ChipStack agentic AI for chip design (4-10x productivity). 80% recurring revenue.	Already have SNPS at 5%. EDA is not a binding physical constraint — it's software. Adding CDNS would over-weight EDA relative to its bottleneck severity.	**NOT RECOMMENDED.** The marginal 1-2% is better deployed into GEV (hard physical bottleneck) or CCJ (structural commodity deficit) than into a second EDA position. SNPS at 5% fully captures the EDA thesis.

Candidate	Ticker	Category	Bull Case	Bear Case	Verdict
Marvell Technology	MRVL	Custom Silicon / Optical Networking	Nvidia invested $2B strategically. Custom silicon business doubled to $1.5B. Revenue guided to $15B by FY28. Silicon photonics for 1.6T transceivers. 20+ AI design wins.	Overlaps with AVGO and ANET. Smaller, lower margins. Doesn't own a new bottleneck.	**NOT RECOMMENDED.** AVGO is the superior custom silicon vehicle. ANET captures networking. Adding MRVL would create overlap without new bottleneck coverage.
Lam Research	LRCX	WFE (Etch/ Deposition)	55% etch market share. Near-100% TSV market for advanced packaging.	Already have ASML (8%) + AMAT (4%) = 12% WFE. Would over-diversify.	**NOT RECOMMENDED.** Portfolio has optimal WFE exposure.
KLA Corporation	KLAC	WFE (Process Control)	Essential for yield management. Growing service revenue.	Same over-diversification concern as LRCX.	**NOT RECOMMENDED.** Same reasoning.
AMD	AMD	AI Accelerators (#2 GPU)	MI300X gaining traction. Credible NVDA alternative.	Not a bottleneck owner. Success captured indirectly via TSMC.	**NOT RECOMMENDED.** Thesis correctly excludes AMD.
ARM Holdings	ARM	Chip Design IP / Royalties	Royalties on every ARM-based chip.	Low-beta royalty model. Not a physical bottleneck.	**NOT RECOMMENDED.** Wrong risk/return profile.
Freeport-McMoRan	FCX	Copper Mining	Essential metal for all DC infrastructure.	Commoditized. Not an oligopoly. Better in Continuity Fund.	**NOT RECOMMENDED.** Too commoditized.
CoreWeave	CRWV	AI Cloud / GPU-as-a-Service	Pure-play AI cloud. Massive GPU fleet.	Buyer of scarce resources, not owner. High debt.	**NOT RECOMMENDED.** Wrong side of bottleneck thesis.
Coherent Corp	COHR	Optical Transceivers (800G/1.6T)	Real bottleneck (20M units needed from 2.5M).	Fragmented market. Not an oligopoly. Thinner margins.	**WATCHLIST.** Real bottleneck but non-oligopolistic market.
NuScale Power	SMR	Small Modular Reactors	First NRC-approved SMR design.	Pre-revenue. Not until ~2030.	**NOT RECOMMENDED.** Too speculative. CEG + CCJ cover nuclear.
Axon Enterprise	AXON	Public Safety / Surveillance AI	$14.4B contracted bookings. 39% revenue growth. AI-native.	Not an AI infrastructure bottleneck. Government budget dependent.	**NOT RECOMMENDED.** Wrong thesis alignment (application, not bottleneck).
Lockheed Martin	LMT	Large Defence Prime	$268B backlog. $900.6B DoD budget.	Not a bottleneck. Low AI beta. Too diversified. In FSKAX already.	**NOT RECOMMENDED.** See C9 analysis above.

Final Recommendations: Stocks to Add

Based on this comprehensive analysis with the Moonshot at 0%, three candidates genuinely fill bottleneck gaps or complete thesis chains:

1. GE Vernova (GEV) — Add at 3% - **Why:** Fills the single largest gap in the portfolio. Power transformers (128-week lead times, only ~3,500 units produced globally/year) and gas turbines (sold out through 2028) are arguably the hardest physical bottleneck in the entire datacenter stack. GEV's $150B backlog and expanding margins (8.5% → 20% by 2028) demonstrate structural pricing power. The $30B electrification backlog provides multi-year revenue visibility. - **Concern:** Stock is expensive (up 107% in 12 months). - **Resolution:** The portfolio thesis is about owning bottleneck companies with structural pricing power, regardless of current valuation. GEV's pricing power will grow as the 175 GW grid shortfall materializes. - **Funding:** From the 5.5% Moonshot redistribution.

2. Cameco (CCJ) — Add at 2% - **Why:** Completes the nuclear thesis chain. CEG owns the reactors (power generation). PWR builds the grid (power delivery). CCJ supplies the fuel and (via Westinghouse) designs and services the reactors. The structural uranium supply deficit (70% of utility requirements through 2045 uncovered) is one of the most compelling commodity imbalances in any market. Westinghouse ownership is a strategic asset. - **Concern:** P/E of 107x. Long-term contracts cap near-term price upside. - **Resolution:** The thesis is multi-year. As the uranium supply deficit widens and nuclear capacity must expand to serve AI power demand, CCJ's contract book and Westinghouse division become increasingly valuable. - **Funding:** From the 5.5% Moonshot redistribution.

3. Core Scientific (CORZ) — Optional add at 1% (deferred) - **Why:** Owns energized grid connections — one of the hardest bottlenecks (3-5 year lead times). The CoreWeave anchor ($10B+ over 12 years) provides revenue visibility. - **Concern:** Transitioning business. Customer concentration. Negative EBITDA. Not a pure bottleneck owner in the same class as GEV or CCJ. - **Resolution:** Defer until LEAPS/options strategies are better understood. CORZ could be an excellent LEAPS candidate for its binary upside. For now, the 5.5% is better deployed into GEV (3%) + CCJ (2%) + a 0.5% increase elsewhere. - **Recommendation:** Do not add to core at this time. Revisit when Moonshot sleeve is reactivated.

Where the Remaining 0.5% Goes

After deploying 3% to GEV, 2% to CCJ, and 1% to CRWD increase (2% → 3%), the remaining 0.5% is absorbed by the INTC trim (5% → 4%) and DELL holding at 5%, keeping the portfolio at exactly 100%.

Part XXIV: Revised Final Allocation (V8.3 — Moonshot at 0%, New Additions)

Sleeve	Ticker	V7.3 (Sept 2025)	V8.3 (March 2026)	Change	Rationale
Energy / Power	CEG	15%	**12%**	**-3%**	Nuclear anchor. Reduced to fund VST; reduces single-stock risk.

Sleeve	Ticker	V7.3 (Sept 2025)	V8.3 (March 2026)	Change	Rationale
	VST (NEW)	0%	**3%**	**+3%**	Gas + nuclear + merchant power. 66% earnings growth. Amazon/Meta PPAs.
	PWR	3%	**3%**	—	Grid transmission bottleneck. Essential enabler.
Foundry	TSM	12%	**12%**	—	Singular chokepoint. 77% advanced node revenue.
WFE	ASML	8%	**8%**	—	100% EUV monopoly. €38.8B backlog.
	AMAT	4%	**4%**	—	Diversified WFE + U.S. geopolitical hedge.
Rack-Scale & Cooling	VRT	8%	**8%**	—	Liquid cooling mandate. $15B backlog.
	DELL	5%	**5%**	—	$43B AI backlog. $50B FY27 guide. Strong execution but not a bottleneck owner.
Accelerators	NVDA	6%	**6%**	—	Training monopoly. $1T Blackwell/Rubin pipeline.
	AVGO	3%	**3%**	—	Custom ASIC enabler. $73B AI backlog.
Memory	MU	6%	**6%**	—	HBM4 sold out. Structural de-risking.
EDA	SNPS	5%	**5%**	—	Toll road on all chip design.
Networking	ANET	5%	**5%**	—	AI Ethernet leader.
Security	CRWD	2%	**3%**	**+1%**	AI-native security. Non-discretionary spend.
Operations	PLTR	2%	**2%**	—	AI operating system / governance layer.
Defense	AVAV	1%	**1%**	—	Small UAS leader. Replicator program.
	KTOS	1%	**1%**	—	Affordable mass / CCA. USMC Valkyrie POR.
Bio-Risk	MRNA	1%	**1%**	—	Tail-risk insurance.
Materials	MP	3%	**3%**	—	Rare earth hedge. Domestic production online.
Domestic Foundry	INTC	5%	**4%**	**-1%**	18A HVM. Apple deal. Taiwan hedge. Volume commitments pending 2H 2026.
Electrical Equipment (NEW)	**GEV**	0%	**3%**	**+3%**	Transformer + gas turbine bottleneck. $150B backlog. 128-week lead times.
Nuclear Fuel (NEW)	**CCJ**	0%	**2%**	**+2%**	Upstream uranium. Structural deficit. 49% Westinghouse.
Moonshot	—	10%	**0%**	**-10%**	Held at 0% pending understanding of LEAPS/ options strategies.
Total		**100%**	**100%**		

Summary of All Changes from V7.3 → V8.3

Change	From	To	Source of Funds	Rationale
INTC to core	0% (Moonshot)	4% core	Moonshot → Core	18A validated. Apple deal. Taiwan hedge. Volume commitments pending 2H 2026.

Change	From	To	Source of Funds	Rationale
GEV (NEW)	0%	3%	Moonshot elimination	Transformer/turbine bottleneck — hardest physical constraint.
CCJ (NEW)	0%	2%	Moonshot elimination	Completes nuclear chain. Structural uranium deficit.
VST (NEW)	0%	3%	CEG reduction	Gas + nuclear + merchant power. Reduces single-stock risk.
CEG reduction	15%	12%	Funds VST addition	15% single-stock concentration was excessive.
CRWD increase	2%	3%	Moonshot elimination	AI security becoming non-discretionary.
Moonshot elimination	10%	0%	N/A	Held at 0% until LEAPS understood. Stocks evaluated as core candidates above.
CORZ/IREN/ APLD	3.5% (Moonshot)	0%	Redirected to GEV/ CCJ/DELL/CRWD	Not pure bottleneck owners. Better as LEAPS candidates when Moonshot reactivated.
KTOS (extra)	2% (Moonshot)	0%	Absorbed into core 1%	Core 1% is sufficient. Extra exposure via LEAPS when ready.
MRNA (extra)	1% (Moonshot)	0%	Absorbed into core 1%	Core 1% is sufficient.

Part XXV: The Event Horizon Fund — Complete Portfolio (V8.3)

A. Portfolio

This portfolio concentrates capital in the hardest, slowest-to-solve physical bottlenecks of the AI buildout (foundry, WFE, power, thermal, electrical equipment, nuclear fuel, and fabric), then rounds out with upstream "blueprints" (EDA), security/governance, strategic materials, and a targeted defense position for asymmetric right-tails. Relative to typical AI exposure, this construction intentionally avoids being the marginal buyer of scarce GPUs and instead owns the entities with structural pricing power when demand collides with physics and policy. The core theses (CoWoS scarcity, HBM oligopoly, 24/7 carbon-free baseload power, diversified gas + nuclear generation, liquid cooling mandate, Ethernet fabric share gains, transformer lead time crisis, uranium supply deficit) remain the spine. Aggregate weights sum to exactly 100% with the Moonshot sleeve held at 0% pending understanding of options strategies. Where new analysis deepened conviction (GEV, CCJ, VST, CRWD), allocations were adjusted accordingly. INTC was trimmed from 5% to 4% (volume commitments pending) and DELL holds at 5% (strong execution but not a bottleneck owner) to maintain precise balance.

Foundry & Advanced Packaging (12%): TSM (12%)

Sleeve Thesis: The physical gatekeeper of AI progress: leading-edge wafer supply plus CoWoS advanced packaging. The forecast shows AI taking the lion's share of ≤5nm by 2029-30, reinforcing TSM's pricing power.

Core Thesis: This sleeve forms the irreplaceable manufacturing foundation of the entire AI compute engine. The demand for SOTA AI chips is forecast to grow at a blistering 80.1% CAGR through 2030. In stark contrast, the supply of leading-edge (≤5nm) wafers is projected to grow at a mere 16.0% CAGR. This vast and persistent divergence between demand and supply is the central, quantifiable imbalance driving the thesis. Furthermore, CoWoS advanced packaging capacity is projected to grow at a 71.3% CAGR — aggressive, but still significantly lagging the 80.1% chip demand growth it must support, ensuring a chronically tight market and sustained pricing power for the sole-source provider. The "Share of <5nm wafer production used for AI chips" is projected to skyrocket from 2.48% in 2023 to 99.48% by 2030 — AI is cannibalizing the entire available supply of advanced manufacturing, crowding out traditional customers (smartphones, PCs) and granting the foundry extraordinary power as the ultimate arbiter of who gets access to the future of silicon.

Conviction Level: Highest. Quantitative proof of structural supply/demand imbalance and industry-wide cannibalization effect.

Top Drivers: Hyperscaler capex cycles; successful ramp of N2/A16 nodes; CoWoS capacity expansion to meet insatiable demand.

Key Risks: Geopolitical disruption in Taiwan; faster-than-expected Intel Foundry Services progress.

TSM (12%): World's largest dedicated semiconductor foundry; manufactures chips for fabless companies like NVDA, AMD, AVGO. Monopoly provider of CoWoS advanced packaging. Q4 2025 revenue: $33.73B (+25.5% YoY). Gross margins: 62.3%. 2026 capex: $52-56B. CoWoS capacity quadrupling to 130,000 wafers/month by late 2026. Advanced Technologies (7nm and below): 77% of revenue. HPC has surpassed smartphones as largest revenue segment (58%). Nvidia alone has secured >60% of 2026 CoWoS capacity for Rubin GPUs.

Why invest: AI-2027's compute explosion is physically impossible without TSM's leading-edge nodes (N3, N2) and CoWoS packaging. This creates a structural bottleneck with immense and durable pricing power, reinforced by competitor execution failures.

What must be true: Must successfully ramp CoWoS capacity to >50,000 wafers/month by end of 2026. Track via TSMC Investor Relations quarterly updates.

Reduce if: Geopolitical conflict in Taiwan escalates materially; Intel's foundry services demonstrate a credible roadmap matching N2 performance and yield by 2027.

AI-2027 Alignment: TSM is the singular chokepoint for production of all advanced AI accelerators — the most critical piece of physical infrastructure in the entire AI stack.

Estimated Returns (Base): 2026: 30-40%, 2027: 25-35%, 2028: 15-25%, 2029: 10-15%, 2030: 10%.

Accelerators & Custom Silicon (9%): NVDA (6%), AVGO (3%)

Sleeve Thesis: The "brains" of the AI revolution. GPU and custom silicon designers capture disproportionate system-level value through their unique ability to integrate scarce HBM and CoWoS-packaged silicon into market-leading systems.

Core Thesis: The AI chip market is projected to grow from $50B in 2023 to over $3T in 2030 (80.1% CAGR). Next-generation SOTA chips are physically doubling in die size each generation (B100:

1,628mm² vs. H100: 814mm²), which exacerbates the global silicon scarcity. The market leader's innovation roadmap directly strengthens its own bottleneck position by consuming a disproportionate share of the world's limited manufacturing capacity.

Conviction Level: High. Immense value capture confirmed with a "compounding scarcity" dynamic.

Top Drivers: Hyperscaler and AGI lab capex cycles; Blackwell/Vera Rubin ramp; CUDA software moat durability.

Key Risks: Significant shift by hyperscalers to custom ASICs for frontier training; severe export controls.

NVDA (6%): Designs GPUs (Blackwell/Vera Rubin), rack-scale systems (GB200 NVL72), networking (NVLink), and the CUDA ecosystem. Q4 FY26 revenue: $68.1B (+73% YoY). Data center revenue: $62.3B (+75% YoY, 91% of total). CEO projects $1T cumulative Blackwell/Rubin demand through 2027. Vera Rubin on track for H2 2026 with 10x inference cost reduction vs. Blackwell.

Why invest: Platform capture across the entire AI stack. By integrating compute, networking, and software, NVIDIA creates a deep moat and captures disproportionate system-level value.

What must be true: Blackwell/Rubin platforms must maintain or grow share against custom ASICs for frontier training.

Reduce if: Two or more hyperscalers significantly shift primary training cluster builds to custom ASICs for two consecutive generations.

AI-2027 Alignment: Primary integrator of scarce CoWoS and HBM supply. The price-setter for AI compute.

Estimated Returns (Base): 2026: 35-48%, 2027: 30-45%, 2028: 20-30%, 2029: 15-20%, 2030: 10-15%.

AVGO (3%): Designs custom ASICs (XPUs) for hyperscalers and high-speed networking silicon. $73B AI product backlog. 100% YoY AI revenue growth. Key design partnerships with top hyperscalers (Google, others).

Why invest: Diversified exposure to AI silicon beyond GPUs. Captures the growing trend of hyperscalers building custom chips to optimize workloads.

What must be true: Must retain key custom silicon partnerships and demonstrate a clear next-gen design-win pipeline.

Reduce if: Its largest custom partner brings chip design fully in-house or shifts to a different design partner.

AI-2027 Alignment: Primary enabler for hyperscalers building their own massive compute fleets.

Estimated Returns (Base): 2026: 25-30%, 2027: 20-25%, 2028: 15-20%, 2029: 10-15%, 2030: 10%.

HBM / Memory (6%): MU (6%)

Sleeve Thesis: HBM is a critical, non-substitutable component for all AI accelerators. The three-player oligopoly (SK Hynix, Samsung, Micron) benefits from both volume growth and immense pricing power.

Core Thesis: HBM production growth (75.8% CAGR) slightly lags SOTA chip demand growth (80.1% CAGR), ensuring a structurally tight market. HBM requirements per chip are rising (6 units in 2023 → 10 by 2029). The "Share of HBM used for AI chips" grows from 80% (2023) to a floor of 98% from 2026 onward, transforming HBM producers from cyclical commodity suppliers to strategic, non-substitutable AI component providers. This structural de-risking should lead to a fundamental re-rating of valuation multiples.

Conviction Level: Upgraded. Supply bottleneck validated plus structural business model de-risking.

Top Drivers: HBM market share gains; pricing power for HBM3E/HBM4; yield execution.

Key Risks: Broader DRAM/NAND cycle downturn; yield risk on new HBM generations.

MU (6%): Designs and manufactures DRAM, NAND, and HBM. Entire 2026 HBM4 capacity sold out through non-cancellable contracts. Q2 2026 revenue: $23.86B. HBM market share: 20-25% (up from mid-teens). Investing $25B in capex for HBM4 expansion. HBM4 delivers 2.3x bandwidth increase over HBM3E with >20% power efficiency, optimized for Vera Rubin. 70% of high-end memory demand now from AI datacenters.

Why invest: Oligopolistic pricing power in a supply-constrained market undergoing structural de-cyclicalization.

What must be true: Must maintain yield and performance parity with SK Hynix on HBM4 generation.

Reduce if: Fails to maintain parity on two consecutive HBM generations (HBM4 + HBM4E).

AI-2027 Alignment: Direct beneficiary of the semiconductor supply bottleneck. Essential ingredient for 10x compute growth.

Estimated Returns (Base): 2026: 40-50%, 2027: 25-40%, 2028: 15-25%, 2029: 10-15%, 2030: 10%.

WFE / Semi Capital Equipment (12%): ASML (8%), AMAT (4%)

Sleeve Thesis: If ≤5nm is ~AI-only by 2030, leading-edge tools remain the capex chokepoint. To relax wafer scarcity, you must first expand the factory that builds factories (EUV, High-NA EUV, etch, deposition, metrology). This is upstream, oligopolistic, and policy-levered.

Core Thesis: The "geopolitical multiplier" drives demand: parallel fab construction in the U.S. (Intel, TSMC Arizona), Europe, Japan, and South Korea all require duplicate equipment sets from the same WFE oligopoly. Government subsidies (CHIPS Act) further accelerate demand.

Top Drivers: EUV/High-NA adoption cycles; government subsidy-driven fab construction; increasing process complexity.

Key Risks: Severe semiconductor capex downturn; tightening export controls.

ASML (8%): World's sole manufacturer of EUV lithography machines — the technology required to produce all advanced AI chips at ≤7nm. Historic $7.9B single order from SK Hynix (March 2026). Q4 2025 bookings: €13.2B (EUV alone: €7.4B, +150%). Total backlog: €38.8B. High-NA EUV transitioning from development to early adoption.

Why invest: An unassailable 100% monopoly on the single most critical technology for advancing semiconductor manufacturing. Without ASML, the AI-2027 compute forecast is physically impossible. Barriers to entry are immense.

What must be true: Must successfully ramp High-NA EUV production and see adoption by leading-edge customers for the 2026+ roadmap.

Reduce if: A credible, non-EUV alternative for 2nm patterning emerges (highly unlikely); severe global restrictions on all EUV exports.

AI-2027 Alignment: Provides the literal light that patterns the silicon heart of the AI revolution. The ultimate gatekeeper of Moore's Law.

Estimated Returns (Base): 2026: 30-40%, 2027: 30-35%, 2028: 20-25%, 2029: 15-20%, 2030: 10-15%.

AMAT (4%): Global leader in wafer fabrication equipment (deposition, etch, inspection). Large U.S. manufacturing footprint makes it a CHIPS Act beneficiary. Stated goal to more than double advanced packaging business to >$3B, aligning with the bottleneck thesis. Provides diversified WFE exposure and U.S.-centric geopolitical hedge against Asia-based risk.

Why invest: "Buy the market" investment in overall WFE spending with a US-centric hedge and advanced packaging upside.

What must be true: Must maintain market share across key segments for 3nm/2nm and convert advanced packaging pipeline into revenue.

AI-2027 Alignment: Provides the essential machinery required to build the world's AI fabs. Captures value from geopolitically-driven supply chain resiliency.

Estimated Returns (Base): 2026: 25-35%, 2027: 20-30%, 2028: 15-20%, 2029: 10-15%, 2030: 10%.

Electronic Design Automation (5%): SNPS (5%)

Sleeve Thesis: The custom silicon multiplier — every hyperscaler tape-out expands EDA demand regardless of who wins GPU share. The stable EDA duopoly owns the "digital blueprints" for the entire semiconductor industry.

Core Thesis: As chip complexity grows exponentially, EDA software becomes increasingly critical, high-margin, and non-substitutable. The rise of custom silicon design by hyperscalers massively expands the TAM. EDA leaders win regardless of who designs the winning AI chip, making them the ultimate arms dealers.

Top Drivers: Custom ASIC design starts; chip complexity growth; AI-powered tool adoption.

Key Risks: Slowdown in semiconductor innovation; open-source EDA disruption (distant).

SNPS (5%): Global leader in EDA software and semiconductor IP. Provides mission-critical tools for designing and verifying complex ICs. Stable duopoly with near-100% customer retention. Highly predictable recurring revenue. Design Automation segment: +23% YoY. AI-powered Synopsys.ai suite driving adoption.

Why invest: High-margin, asset-light software monopoly that benefits from every semiconductor growth trend without taking on hardware capital intensity or competitive risk.

What must be true: Must maintain market share leadership and demonstrate continued innovation of AI-powered tool suite.

Reduce if: Loses significant share to Cadence for two consecutive years, or if operating margins materially decline.

AI-2027 Alignment: Essential digital nervous system for the entire AI silicon ecosystem. Captures a high-margin toll on every chip designed.

Estimated Returns (Base): 2026: 25-30%, 2027: 25-30%, 2028: 20-25%, 2029: 15-20%, 2030: 10-15%.

AI Data Center Fabric (5%): ANET (5%)

Sleeve Thesis: As training scales, east-west traffic and low-latency fabrics become non-discretionary. The shift from proprietary InfiniBand to open Ethernet is a durable, multi-year trend in early innings.

Core Thesis: The network fabric is a distinct, critical bottleneck responsible for connecting tens of thousands of GPUs into a single, coherent supercomputer. Overwhelming evidence indicates Ethernet is rapidly displacing InfiniBand for large-scale AI deployments.

Top Drivers: 400G/800G adoption; AI cluster buildout velocity; Ethernet market share gains vs. InfiniBand.

Key Risks: Nvidia Spectrum-X competition; hyperscaler in-house networking efforts.

ANET (5%): Undisputed technology and market share leader in high-performance cloud networking for AI clusters. Leadership in 400G/800G Ethernet switches combined with EOS software. Management projects >$1.5B in aggregated AI networking revenue in 2025. Deeply embedded in the networks of the world's largest cloud providers.

Why invest: Premier pure-play investment on the AI datacenter fabric buildout.

What must be true: Must demonstrate continued market share gains in 400G/800G switching for AI cluster deployments while defending gross margins.

Reduce if: Nvidia Spectrum-X consistently wins the majority of new large-scale AI cluster networking contracts for two consecutive quarters.

AI-2027 Alignment: The central nervous system of the AI factory. Builds the ultra-low-latency Ethernet fabric allowing tens of thousands of GPUs to function as one system.

Estimated Returns (Base): 2026: 25-35%, 2027: 25-35%, 2028: 15-25%, 2029: 10-15%, 2030: 10%.

Rack-Scale Integration & Liquid Cooling (13%): VRT (8%), DELL (5%)

Sleeve Thesis: The power density of next-generation AI accelerators makes traditional air cooling obsolete. Rack power densities exceeding 100-200 kW make liquid cooling a physical requirement. The "product" is no longer a server but a fully integrated, liquid-cooled rack-scale system.

Core Thesis: Liquid cooling becomes mandatory, high-margin enabling technology. The market matures from "speed-to-market" to "scale-of-deployment," favoring leaders with global supply chains and deep enterprise relationships.

Top Drivers: Direct-to-chip liquid cooling adoption; GPU power density trends; large-scale integrator backlog conversion.

Key Risks: New cooling technologies; supply chain issues for specialized cooling components.

VRT (8%): Provides critical data center infrastructure specializing in power and thermal management, including market-leading direct-to-chip liquid cooling. Organic orders growth: 252% YoY in Q4 2025 — strongest quarter in company history. Backlog: $15B (+109% YoY). Book-to-bill ratio: 2.9x. Stock up ~64% YTD in 2026, ~200% over past year.

Why invest: Extreme power density of AI accelerators makes advanced liquid cooling mandatory. VRT is the market leader aggressively expanding manufacturing capacity with critical design wins establishing powerful incumbency.

What must be true: At least two of the top three hyperscalers must announce >50% of new AI clusters will be direct-to-chip liquid cooled by 2027.

Reduce if: A competitor (Schneider Electric) consistently wins majority of new large-scale cooling deployments for two+ quarters.

AI-2027 Alignment: Solves the thermal bottleneck — a direct physical consequence of massive compute density.

Estimated Returns (Base): 2026: 50-65%, 2027: 35-45%, 2028: 25-30%, 2029: 15-20%, 2030: 10-15%.

DELL (5%): Global leader in IT solutions. Leveraging 58% server market share and massive enterprise salesforce to dominate the scaled AI deployment phase. FY2026: $24.7B in AI server revenue (vs. $20B target). FY2027 guide: $50B AI server revenue (+103% YoY). AI backlog: $43B. 4,000+ AI server customers. Massively outscales competitors.

Why invest: Primary integrator translating silicon into tens of thousands of deployable rack-scale systems. Captures value at the system level through scale and logistics.

What must be true: Must convert AI server backlog to revenue, achieving >$40B in AI server shipments in FY2027 while maintaining ISG margins above 9%.

Reduce if: ISG revenue growth decelerates for two consecutive quarters or margins fall significantly below 9%.

AI-2027 Alignment: Primary integrator for the global AI factory buildout. Captures system-level value.

Estimated Returns (Base): 2026: 45-55%, 2027: 30-40%, 2028: 20-25%, 2029: 10-15%, 2030: 10%.

Energy / Power Generation (20%): CEG (12%), VST (3%), PWR (3%), CCJ (2%)

Sleeve Thesis: As the AI buildout progresses, the primary systemic bottleneck shifts from silicon to megawatts. The resulting structural shock to the energy grid triggers a fundamental repricing of existing power assets from commodity assets to irreplaceable strategic infrastructure. This sleeve now captures

the full power chain: nuclear-first generation (CEG), diversified gas + nuclear generation (VST), grid delivery (PWR), and uranium fuel (CCJ).

Core Thesis: "Total AI datacenter power demand" exhibits the highest growth rate of any metric analyzed: 107.5% CAGR projected from 2023 to 2030. The forecast projects 77 GW by end of 2027, with 134 GW by 2030 and a potential 175 GW shortfall by 2033. Hyperscalers are pursuing an "all of the above" power strategy: nuclear for premium 24/7 zero-carbon baseload, natural gas for near-term dispatchable capacity, and renewables as supplement. The portfolio mirrors this strategy with CEG (nuclear anchor) and VST (gas + nuclear complement). Power is the hardest and most durable bottleneck in the entire AI stack.

Conviction Level: Highest in the portfolio. Expanded to 20% with the addition of CCJ and VST. The CEG/VST split reduces single-stock risk while maintaining maximum power exposure.

Top Drivers: PPA pricing and duration; nuclear policy support; electricity demand growth in key datacenter markets; uranium supply deficit.

Key Risks: Unplanned nuclear outages; adverse regulatory changes (PJM price caps); Calpine integration execution; breakthrough in long-duration storage.

Addressing the Q1 2026 Power Stock Decline (~22-32% from peaks):

The entire power/AI infrastructure trade compressed in Q1 2026: CEG fell from its $413 all-time high (Oct 15, 2025) to $279 (Mar 31, 2026) — a 32% decline. VST dropped 31% from its $220 high. TLN fell 29% from $451. Four distinct headwinds converged:

1. **PJM Price Cap Scare (Jan 16 — CEG dropped 9.8% in one day):** The Trump administration and Mid-Atlantic governors proposed caps on wholesale electricity rates to prevent "price gouging" by IPPs, threatening the nuclear premium pricing thesis. However, PJM capacity auctions are governed by federal tariff rules under FERC jurisdiction — wholesale electricity markets are federally regulated, and neither state governors nor the President can unilaterally impose wholesale price caps. No caps have been implemented. Bilateral PPAs (Microsoft, Meta) are private contracts not subject to wholesale market interventions.
2. **FERC Rejection of Behind-the-Meter Nuclear Deals:** FERC blocked the original Amazon-Susquehanna co-location structure. This forced a restructuring to front-of-meter delivery, adding regulatory friction. However, the restructured deal (Talen-Amazon, $18B) was approved, establishing the template for future nuclear-to-datacenter arrangements.
3. **CEG 2026 Guidance Miss (Mar 31 — dropped 6.5%):** Guided $11-12 EPS vs. $11.72 consensus. The miss reflected elevated 2026 nuclear outage schedule and Calpine integration costs. However, this still represents ~30%+ earnings growth, and CEG simultaneously announced a $5B share buyback — the opposite signal of a company in distress.
4. **Broader Market Risk-Off:** Tariff escalation, Iran nuclear deadline, and general AI infrastructure valuation compression hit all power names simultaneously. This is macro, not company-specific.

Assessment: The decline is driven by regulatory headline risk, integration noise, and macro sentiment — not by deterioration in the underlying demand curve. AI datacenter power demand grew from 50.5 GW (2024) → 61.8 GW (2025) → 75.8 GW (2026), and hyperscalers are signing more nuclear PPAs, not fewer. The one legitimate risk to monitor is regulatory action on wholesale pricing — if PJM or FERC

actually caps PPA pricing, the nuclear premium thesis takes a structural hit. The CEG/VST split partially hedges this: VST's diversified gas + nuclear + merchant model is more resilient to nuclear-specific regulatory risk than a pure-nuclear play.

CEG (12%): Largest producer of carbon-free energy in the U.S. Operating the nation's largest fleet of nuclear power plants (32 GW, 21 reactors, 12 facilities). Completed $16.4B Calpine acquisition (Jan 2026), making it the largest electricity producer in the U.S. with ~55-60 GW capacity. Now offers blended nuclear + gas + geothermal power products. Microsoft PPA at $110-115/MWh (2-3x wholesale). Meta 20-year nuclear PPA. TMI restart proceeding. $840M GSA contract (largest in GSA history). Down 32% from Oct 2025 ATH ($413 → $279) — a buying opportunity driven by regulatory headline risk and integration noise, not thesis deterioration. Calpine integration requires $5B divestiture of 4.4 GW gas capacity to LS Power per DOJ antitrust resolution. 2026 guided EPS: $11-12 (~30% growth). $5B share buyback announced. Analyst consensus target: ~$394 (41% upside from $279).

Why invest: CEG's 32 GW nuclear fleet is the single most irreplaceable asset class in the AI power chain. Nuclear plants take 10-15 years to build; CEG owns the majority of what exists. Premium PPA pricing ($100-115/MWh) reflects this irreplaceability. The 32% drawdown from ATH creates an attractive entry point — the stock is cheaper now than at any point since June 2025, while the underlying demand curve has only steepened.

Why reduced from 15% to 12%: 15% in a single stock was the portfolio's largest concentration risk. The Calpine integration introduces execution risk (antitrust divestitures, elevated interest expense, integration complexity). PJM price cap headline risk adds regulatory uncertainty specific to CEG's nuclear-heavy fleet. Splitting 3% to VST provides diversification within the power thesis while maintaining 15% total nuclear + gas generation exposure and partially hedging against nuclear-specific regulatory risk.

What must be true: Must secure at least one new PPA >500 MW with a major tech company at >$90/MWh by mid-2026.

Reduce if: Major breakthrough in long-duration energy storage (>100-hour batteries) becomes commercially viable before 2028; or if Calpine integration materially underperforms (>20% accretion target missed).

AI-2027 Alignment: Provides the solution to the final, hardest bottleneck: reliable, large-scale power generation.

Estimated Returns (Base): 2026: 35-50% (rebound from YTD weakness), 2027: 40-50%, 2028: 25-35%, 2029: 20-25%, 2030: 15-20%.

VST (3%): (NEW) Largest independent power producer in the U.S. with ~41 GW total capacity. Diversified fleet: ~20 GW combined-cycle gas turbines, 6.4 GW nuclear (Comanche Peak + Energy Harbor fleet), ~1.2 GW battery storage. Acquiring 5,500 MW Cogentrix gas capacity (closing mid-late 2026). Amazon PPA: 1,200 MW, 20-year at Comanche Peak nuclear. Meta PPA: 2,600 MW, 20-year across PJM nuclear facilities. 2025 EBITDA: $5.9B (beat guidance). 2026 EBITDA guided: $6.8-7.6B (22% growth). 2026 earnings growth: ~66%. Completed $5.6B in share repurchases since 2021, another $1B authorized. Targeting $10B+ cash generation through year-end 2027. Stock at $150, down ~31% from $220 52-week high. Analyst consensus target: ~$237 (58% upside). Beta: 1.44.

Why invest: VST complements CEG by providing the "all of the above" power exposure that mirrors how hyperscalers are actually buying power. Its gas fleet captures near-term wholesale electricity price spikes (critical during the current oil crisis), while its nuclear fleet secures long-term premium PPAs. Faster 2026 earnings growth (66% vs. CEG's 24%) provides more near-term appreciation potential. Higher beta (1.44) gives more leverage to the power thesis.

What must be true: Must successfully close the Cogentrix acquisition and demonstrate EBITDA within the $6.8-7.6B guidance range. Nuclear PPAs (Amazon, Meta) must proceed on schedule.

Reduce if: Wholesale electricity prices collapse (would hurt merchant power revenue); or if nuclear plant operational issues arise at Comanche Peak or Energy Harbor facilities.

AI-2027 Alignment: VST is the "flexibility partner" for hyperscalers — providing bundled gas, nuclear, and battery solutions that deliver reliable power across all market conditions.

Estimated Returns (Base): 2026: 40-60% (rebound from discount + earnings growth), 2027: 35-50%, 2028: 25-35%, 2029: 15-25%, 2030: 10-15%.

PWR (3%): Market leader in high-voltage transmission lines and substations. Physically builds the grid infrastructure required to deliver power to datacenters. Record $35.8B backlog reflects immense demand for specialized electrical construction. Grid interconnection queues are jammed — 4-year transformer lead times, 5-7 year interconnection waits in key markets.

Why invest: The electrical grid itself is a co-equal Tier 1 bottleneck alongside generation. Investing only in power generation without owning the means of delivery is an incomplete thesis.

What must be true: Must maintain or grow backlog as grid modernization spending accelerates.

Reduce if: Grid infrastructure spending decelerates meaningfully due to regulatory or political headwinds.

AI-2027 Alignment: Builds the physical delivery infrastructure for the power that the AI revolution requires.

Estimated Returns (Base): 2026: 20-30%, 2027: 25-35%, 2028: 20-25%, 2029: 15-20%, 2030: 10-15%.

CCJ (2%): (NEW) The only publicly traded, fully integrated uranium producer with majority ownership of a reactor design and services company. Mining (McArthur River — world's highest-grade uranium mine), fuel services, and 49% Westinghouse (AP1000 reactor design, AP300 SMR, reactor servicing, $370-430M EBITDA in 2026). Structural uranium supply deficit: 190M lbs demand vs. 135M lbs mine supply. 70% of utility uranium requirements through 2045 remain uncovered. Spot uranium at ~$82/lb, long-term contracts approaching $100/lb. Deliberate supply discipline: producing 19.5-21.5M lbs at McArthur River vs. 25M lb licensed capacity. U.S. government committed $80B for new reactor construction and added uranium to critical minerals list. $2.6B India supply agreement.

Why invest: Completes the nuclear thesis chain: CEG owns and operates the reactors (power generation), PWR builds the grid (power delivery), CCJ supplies the fuel and (via Westinghouse) designs and services the reactors. The structural uranium supply deficit is one of the most compelling commodity imbalances in any market. Westinghouse ownership provides strategic optionality for SMR deployment.

What must be true: Global nuclear capacity expansion must continue (438 GW today → 746 GW by 2040), with no major political reversal on nuclear energy support.

Reduce if: A major nuclear policy reversal occurs in 2+ G7 countries; or if uranium supply unexpectedly surges (new major deposits or substitution breakthrough).

AI-2027 Alignment: Provides the fuel for the nuclear plants that power the AI revolution. Westinghouse ownership gives exposure to next-generation reactor design.

Estimated Returns (Base): 2026: 15-25%, 2027: 25-35%, 2028: 30-40%, 2029: 20-30%, 2030: 15-20%.

Electrical Equipment (3%): GEV (3%)

Sleeve Thesis: (NEW) Power transformers and gas turbines are the single hardest physical bottleneck in datacenter construction. Transformers have 128-week (2.5+ year) lead times with only ~3,500 large power transformers produced globally per year. Gas turbines are sold out through 2028. No other portfolio position captures this critical layer.

Core Thesis: GE Vernova is the dominant manufacturer of both gas turbines (the near-term workhorse for AI datacenter power, providing 40%+ of current supply) and power transformers (the binding constraint on grid expansion). Every gigawatt of new datacenter capacity requires $200-300M in GEV electrical equipment. The company's $150B total backlog, expanding margins (8.5% → 20% by 2028), and the growing physical impossibility of the grid keeping up with demand create structural pricing power on par with ASML and TSMC.

Top Drivers: Gas turbine orders (HA-class demand for baseload); transformer backlog growth; electrification segment expansion (HVDC, substations); margin expansion as slot reservation agreements monetize.

Key Risks: Wind segment drag on margins; macro slowdown reducing utility capex; potential for transformer production to scale faster than expected (would reduce scarcity premium).

GEV (3%): Total backlog: ~$150B. Gas turbine backlog: 83 GW (up from 62 GW in Q4 2025), targeting 100 GW by end of 2026. Slot reservation agreements: 43 GW (20-25% upfront deposits). Gas turbine orders: +77% organically in Q4 2025. Electrification backlog: $30B, expected to double by 2028. Ramping capacity to ~20 GW/year of turbines by mid-2026. EBITDA margins expanding from 8.5% to 20% by 2028. Stock up ~107% in 12 months. BofA target: $3,050.

Why invest: GEV controls two of the hardest bottlenecks in the entire AI infrastructure stack simultaneously — transformers and gas turbines. No other single company provides this dual bottleneck exposure. The 128-week transformer lead time means current demand is physically impossible to fulfill quickly, creating durable pricing power for years.

What must be true: Gas turbine order growth must remain positive as hyperscalers and utilities continue to build out datacenter power capacity. Transformer backlog must continue to grow.

Reduce if: Transformer production globally scales dramatically faster than expected (e.g., new factories online within 2 years vs. the expected 4-5 years); or if the grid buildout slows materially due to regulatory obstruction.

AI-2027 Alignment: GEV manufactures the physical equipment (turbines + transformers) that converts fuel into electricity and delivers it to the datacenter. Without GEV's equipment, no new datacenter gets power.

Estimated Returns (Base): 2026: 25-35%, 2027: 35-45%, 2028: 25-35%, 2029: 15-20%, 2030: 10-15%.

AI-Native Security (3%): CRWD (3%)

Sleeve Thesis: The AI-2027 scenario forecasts superhuman hacking capabilities and model subversion risks. A new security paradigm is required — one that uses AI to defend against AI.

Core Thesis: Spending on AI security is becoming non-discretionary for any organization deploying frontier models. The winning platform uses its own AI to defend against malicious AI — the only logical and scalable response to exponentially accelerating hacking capabilities.

Top Drivers: High-profile AI security breaches; regulatory mandates; platform consolidation.

Key Risks: Competition from security built into AI platforms (NVIDIA, Microsoft); slow adoption.

CRWD (3%): Cloud-native cybersecurity platform focused on endpoints, cloud workloads, identity, and data. Accelerating ARR growth: 23% YoY. Net new ARR growth: 73% in fiscal Q3 2026. AI-SPM (Security Posture Management) and agent-native security solutions directly target AI-2027 threat landscape. Strategy of using AI to defend against AI is the most logical approach. Identity protection module is one of its fastest-growing products.

Why invest: Best positioned of major security platforms to address the new threat landscape created by AI. Its AI-SPM and agentic security solutions are a precise fit for the risks of model subversion and theft.

What must be true: Must demonstrate continued ARR growth >20% YoY and significant AI-SPM module adoption contributing to net new ARR by fiscal 2027.

Reduce if: ARR growth falls below 20% for two consecutive quarters; or if AI-focused modules fail to gain meaningful adoption by mid-2026.

AI-2027 Alignment: Addresses the critical security risks (subversion, theft) that are an inevitable consequence of powerful, autonomous AI.

Estimated Returns (Base): 2026: 25-35%, 2027: 30-40%, 2028: 25-30%, 2029: 20-25%, 2030: 15-20%.

AI-Native Operate & Control (2%): PLTR (2%)

Sleeve Thesis: As AI agents become more powerful, the "operate and control" layer becomes a critical governance bottleneck. Palantir's AIP platform provides the most integrated AI-native "operating system" for managing, auditing, and sandboxing agentic workflows in high-stakes environments.

PLTR (2%): Palantir's AIP platform. U.S. commercial revenue growth: 121% YoY in late 2025. AIP (Artificial Intelligence Platform) gaining rapid enterprise adoption as organizations need a governance/orchestration layer for AI agent deployment. Ontology provides "safety rails" preventing hallucinations

and errors in production. Serves both commercial and government customers. Premium valuation reflects first-mover advantage in a critical, emerging category.

Why invest: Best-positioned company for the governed agentic AI operating system, especially in high-stakes deployments (defense, healthcare, financial services).

What must be true: Must demonstrate sustained profitability and expanding AIP adoption beyond early adopters.

Reduce if: Competitive offerings from established enterprise platforms (ServiceNow, Microsoft) capture significant share of the AI governance market.

AI-2027 Alignment: Provides the governance layer that enables safe deployment of increasingly autonomous AI agents.

Estimated Returns (Base): 2026: 20-30%, 2027: 25-40%, 2028: 20-30%, 2029: 15-20%, 2030: 10-15%.

AI-Enabled Military (2%): AVAV (1%), KTOS (1%)

Sleeve Thesis: The AI-2027 fast takeoff is a geopolitical race. The "affordable mass" doctrine — shifting from small numbers of expensive platforms to large numbers of cheap, autonomous systems — is the most AI-aligned defense trend. These two companies are pure-play on this transformative shift.

Top Drivers: DoD budget allocations for unmanned systems; CCA program wins; AI/autonomy integration into fielded systems.

Key Risks: Defense budget cuts; program delays; competition from large primes.

AVAV (1%): Global leader in small UAS and loitering munitions. AV_Halo platform is a strategic attempt to build the unifying command and control layer for the future battlefield. Selected for DoD Replicator initiative (Tranche 1) with Switchblade 600.

Why invest: Incumbent small UAS leader with a strategically sound vision for becoming the dominant AI/software provider for autonomous systems.

What must be true: Must demonstrate successful AV_Halo adoption beyond its own hardware and secure major P550 UAS program-of-record.

Reduce if: Fails to transition business model toward software/services; margins remain compressed from hardware competition.

AI-2027 Alignment: Distributed eyes, ears, and unifying software brain for the AI-enabled battlefield.

Estimated Returns (Base): 2026: 25-35%, 2027: 35-45%, 2028: 25-30%, 2029: 15-20%, 2030: 15%.

KTOS (1%): Pioneer in high-performance, jet-powered unmanned tactical aircraft designed to be attritable. XQ-58 Valkyrie "loyal wingman." $231M USMC contract with Northrop Grumman. Focus on low-cost, scalable manufacturing of autonomous combat aircraft.

Why invest: Pure-play on the "affordable mass" paradigm central to Pentagon strategy for near-peer competition.

What must be true: XQ-58 Valkyrie or derivative must be selected as a key CCA platform.

Reduce if: Loses out on major CCA program tranches to Boeing or General Atomics.

AI-2027 Alignment: Builds the expendable, autonomous swarms defining future air combat.

Estimated Returns (Base): 2026: 25-35%, 2027: 35-45%, 2028: 25-30%, 2029: 15-20%, 2030: 15%.

Domestic Foundry (4%): INTC (4%)

Sleeve Thesis: Intel's foundry turnaround serves dual purpose: offensive upside (if 18A succeeds and attracts external customers) and defensive hedge (if Taiwan contingency materializes, Intel becomes the most strategically important company on Earth).

INTC (4%): 18A node at HVM with 55-65% yields, targeting 75%+ by early 2027. Apple signed foundry agreement for entry-level M-series chips. Nvidia in "advanced discussions" for foundry use starting 2028. U.S. government holds ~10% equity via CHIPS Act conversions. RibbonFET and PowerVia provide 6-12 month technology lead in backside power delivery. Trimmed from 5% to 4% because external volume commitments are not expected until 2H 2026 — the most speculative core position.

Why invest: The only viable Western alternative to TSMC for leading-edge foundry. Government backing creates a "sovereign floor" under the stock. Each external design win (Apple, potentially Nvidia) is a massive de-risking event.

What must be true: Must achieve 75%+ yields on 18A by early 2027 and convert at least one more major external customer (Nvidia, Qualcomm, or AMD).

Reduce if: 18A yield progression stalls below 65% for two consecutive quarters; or if Apple cancels or scales back its foundry agreement.

AI-2027 Alignment: If Taiwan is disrupted, Intel becomes the sole remaining source of leading-edge logic capacity for the Western world. The insurance value alone justifies the position.

Estimated Returns (Base): 2026: 40-60%, 2027: 30-50%, 2028: 20-30%, 2029: 15-20%, 2030: 10-15%.

Bio-Risk Insurance (1%): MRNA (1%)

Sleeve Thesis: AI advancements create novel, potentially catastrophic bio-risks. This is insurance against one of the most severe negative externalities of rapid AGI takeoff.

MRNA (1%): Pioneering mRNA therapeutics and vaccines. AI-native, digital-first platform enabling unparalleled speed (63 days from sequence to clinical vaccine during COVID). True platform technology addressing any protein-based pathogen. In a tail-risk pandemic scenario, value would experience explosive, non-linear repricing.

Why invest: "Black swan" hedge. In the base case, treads water on its non-COVID pipeline. In a tail event, provides 5-20x asymmetric upside. At 1%, cost of carry is negligible.

What must be true: Must continue integrating AI into discovery/manufacturing to maintain speed advantage.

Reduce if: Not intended for long-term holding in base case. Re-evaluate annually based on biological threat level.

AI-2027 Alignment: Critical hedge against one of the most dangerous consequences of unconstrained AI proliferation.

Estimated Returns (Base): 2026: 10-15%, 2027: 10-15%, 2028: 10%, 2029: 10%, 2030: 10%. (Event-driven: 5-20x in tail scenario.)

Strategic Materials (3%): MP (3%)

Sleeve Thesis: The entire AI infrastructure buildout rests on a vulnerable critical minerals supply chain. The U.S. is almost entirely import-reliant on China for non-substitutable materials (gallium: 98%, germanium: 60%, rare earths: 87% of refining). This is a "kill switch" risk that must be hedged.

MP (3%): America's only fully integrated rare earth producer. Mining, processing (Mountain Pass, CA), metallization, and magnet manufacturing (Fort Worth, TX). Multi-billion dollar DoD partnership including $400M equity investment, 10-year magnet offtake agreement, and 10-year NdPr oxide price floor. Commercial partnerships with Apple and GM. Fort Worth facility now in commercial production.

Why invest: Designated "national champion" with government backstop. The only company creating a fully domestic rare earth magnet supply chain. China's continued weaponization of critical mineral exports validates the hedge thesis.

What must be true: Must successfully ramp Fort Worth magnetics facility to commercial scale meeting quality standards.

Reduce if: China reverses its mineral export restrictions (removing the hedge rationale); or if MP fails to meet quality/volume targets for key customers.

AI-2027 Alignment: Provides foundational, non-substitutable rare earth materials for AI-enabled defense systems and advanced electronics. Cornerstone of U.S. technological independence.

Estimated Returns (Base): 2026: 15-25%, 2027: 20-30%, 2028: 20-25%, 2029: 15-20%, 2030: 10-15%.

Moonshot Sleeve (0%): HELD AT ZERO

The Moonshot sleeve is held at 0% allocation pending a deeper understanding of LEAPS options strategies, theta decay mechanics, and defined-risk options structures. The 10% previously allocated has been redistributed as follows:

Former Moonshot Position	Disposition
INTC (3.5%)	Promoted to core at 4%
CORZ/IREN/APLD (3.5%)	Not added to core — better as LEAPS candidates when Moonshot is reactivated
KTOS additional (2.0%)	Core 1% is sufficient; extra exposure via LEAPS when ready
MRNA additional (1.0%)	Core 1% is sufficient; extra exposure via LEAPS when ready

When to Reactivate: Once LEAPS debit call spreads, expiration management, and roll discipline are understood, the Moonshot can be reactivated at up to 10% with the following recommended candidates: - **INTC:** "Geopolitical champion" LEAPS — Jan '29 debit call spreads targeting 3-6x on

turnaround re-rate - **CORZ/IREN/APLD:** AI landlord LEAPS — convexity via options on the crypto-to-AI transition - **KTOS:** CCA binary catalyst LEAPS — all-options structure for defense program awards - **MRNA:** Bio-risk tail LEAPS — lotto ladder for pandemic tail events

These positions are not in the portfolio now but are flagged as the highest-conviction LEAPS candidates for when the Moonshot sleeve is reactivated.

Projected Annual Returns by Sleeve (Updated for V8.3)

Sleeve	Weight	2026E	2027E	2028E
Energy (CEG/VST/PWR/CCJ)	20%	+28-45%	+35-50%	+25-35%
Foundry (TSM)	12%	+30-40%	+25-35%	+15-25%
WFE (ASML/AMAT)	12%	+28-38%	+25-35%	+18-25%
Rack-Scale (VRT/DELL)	13%	+45-60%	+30-40%	+20-25%
Accelerators (NVDA/AVGO)	9%	+30-42%	+28-40%	+18-28%
Memory (MU)	6%	+40-50%	+25-40%	+15-25%
EDA (SNPS)	5%	+25-30%	+25-30%	+20-25%
Networking (ANET)	5%	+25-35%	+25-35%	+15-25%
Electrical Equipment (GEV)	3%	+25-35%	+35-45%	+25-35%
Security (CRWD)	3%	+25-35%	+30-40%	+25-30%
Domestic Foundry (INTC)	4%	+40-60%	+30-50%	+20-30%
Other (PLTR/AVAV/KTOS/MRNA/MP)	8%	+15-30%	+25-40%	+15-25%
Weighted Portfolio	**100%**	**~32-42%**	**~30-42%**	**~20-28%**

Cumulative 3-year return (base case): ~110-150% (roughly 2-2.5x) **Cumulative 3-year return (bull case):** ~170-230% (roughly 2.7-3.3x) **Cumulative 3-year return (with reactivated Moonshot hitting):** ~220-300% (roughly 3.2-4x)

Final recommendation: Deploy. Hold. Monitor quarterly against the "must-be-true" thresholds. Rebalance only on thesis invalidation, not on price volatility. Reactivate the Moonshot sleeve once LEAPS strategies are understood.

This analysis was prepared on March 31, 2026, incorporating all available data through that date. It synthesizes the Event Horizon Fund Draft V7.3 (September 2025), GPT5 and Gemini analyses (September 2025 and January 2026), the AI-2027 scenario forecast, Situational Awareness (Aschenbrenner, June 2024), METR Long Horizon Tasks research (March 2025), and Epoch AI Trends data.